Mathematics II

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You can find the last version of these course materials at

https://mbujosab.github.io/MatematicasII/

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Part IV
Ortogonality

LECTURE 11: Orthogonal vectors and subspaces

Lecture 11

(**Lecture** 11) S-1 Highlights of Lesson 11

Highlights of *Lesson 11*

- Orthogonal vectors and subspaces
- Null space $\perp$ row space

$$\mathcal{N}(\mathbf{A}) \perp \mathcal{C}(\mathbf{A}^{\mathsf{T}})$$

- Left null space $\perp$ column space

$$\mathcal{N}(\mathbf{A}^{\mathsf{T}}) \perp \mathcal{C}(\mathbf{A})$$

F1

(**Lecture** 11) S-2 Some definitions

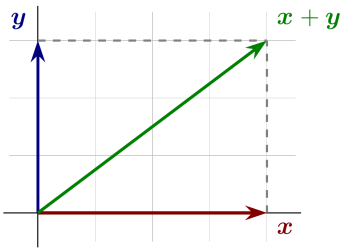
- Dot product

$$\boldsymbol{a} \cdot \boldsymbol{b} = \sum_{i=1}^{n} a_i b_i$$

- Length of a vector $\|\boldsymbol{a}\| = \sqrt{\boldsymbol{a} \cdot \boldsymbol{a}}$ $\qquad \boldsymbol{a} \cdot \boldsymbol{a} = \|\boldsymbol{a}\|^2.$
- Unit vector: $\|\boldsymbol{a}\| = 1$ $\qquad \frac{1}{\|\boldsymbol{x}\|} \cdot \boldsymbol{x}$
- Orthogonal (perpendicular) vectors: $\boldsymbol{x} \cdot \boldsymbol{y} = 0$.

F2

(**Lecture** 11) **S-3** **Orthogonal vectors**



$$\boldsymbol{x} \cdot \boldsymbol{y} \;=\; 0 \quad \Longleftrightarrow \quad \boldsymbol{x} \perp \boldsymbol{y}$$

Pythagorean Thm.: $\boldsymbol{x} \cdot \boldsymbol{y} \;=\; 0 \quad \Longleftrightarrow \quad \|\boldsymbol{x}\|^2 + \|\boldsymbol{y}\|^2 = \|\boldsymbol{x}+\boldsymbol{y}\|^2$

$$\boldsymbol{x}\cdot\boldsymbol{x} + \boldsymbol{y}\cdot\boldsymbol{y} = (\boldsymbol{x}+\boldsymbol{y})\cdot(\boldsymbol{x}+\boldsymbol{y}).$$

F3

(**Lecture** 11) **S-4** **Squared length of a vector**

$$\|\boldsymbol{v}\|^2 = \boldsymbol{v}\cdot\boldsymbol{v}$$

$$\boldsymbol{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad \rightarrow \quad \|\boldsymbol{x}\|^2 = \qquad ; \qquad \boldsymbol{y} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} \quad \rightarrow \quad \|\boldsymbol{y}\|^2 = \qquad ;$$

Are these vectors orthogonal?

$$\boldsymbol{x}+\boldsymbol{y} = \begin{pmatrix} \quad \\ \quad \\ \quad \end{pmatrix}; \qquad \|\boldsymbol{x}+\boldsymbol{y}\|^2 = \qquad ;$$

(Pythagorean T.) $\boldsymbol{x}\cdot\boldsymbol{x} + \boldsymbol{y}\cdot\boldsymbol{y} = (\boldsymbol{x}+\boldsymbol{y})\cdot(\boldsymbol{x}+\boldsymbol{y})$ $\Longleftrightarrow$ (Orthogonality) $\boldsymbol{x}\cdot\boldsymbol{y} = 0.$

F4

(**Lecture** 11) **S-5** **Orthogonal subspaces**

When the subspace $\mathcal{S}$ is orthogonal to subspace $\mathcal{T}$:

Every vector in $\mathcal{S}$ is orthogonal to every vector in $\mathcal{T}$

Are the plane of the *blackboard* and the floor orthogonal?

F5

(**Lecture** 11) S-6 **Null space orthogonal to row space**

- $\mathcal{N}\left(\mathbf{A}\right) \perp$ rows of $\mathbf{A}$

$$\mathbf{A}\boldsymbol{x} = \mathbf{0} \quad \Longrightarrow \quad \begin{pmatrix} ({}_1|\mathbf{A}) \cdot \boldsymbol{x} \\ \vdots \\ ({}_m|\mathbf{A}) \cdot \boldsymbol{x} \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

- $\mathcal{N}\left(\mathbf{A}\right) \perp \boldsymbol{d}\mathbf{A}, \quad \forall \boldsymbol{d} \in \mathbb{R}^m$ (any linear combination of the rows)

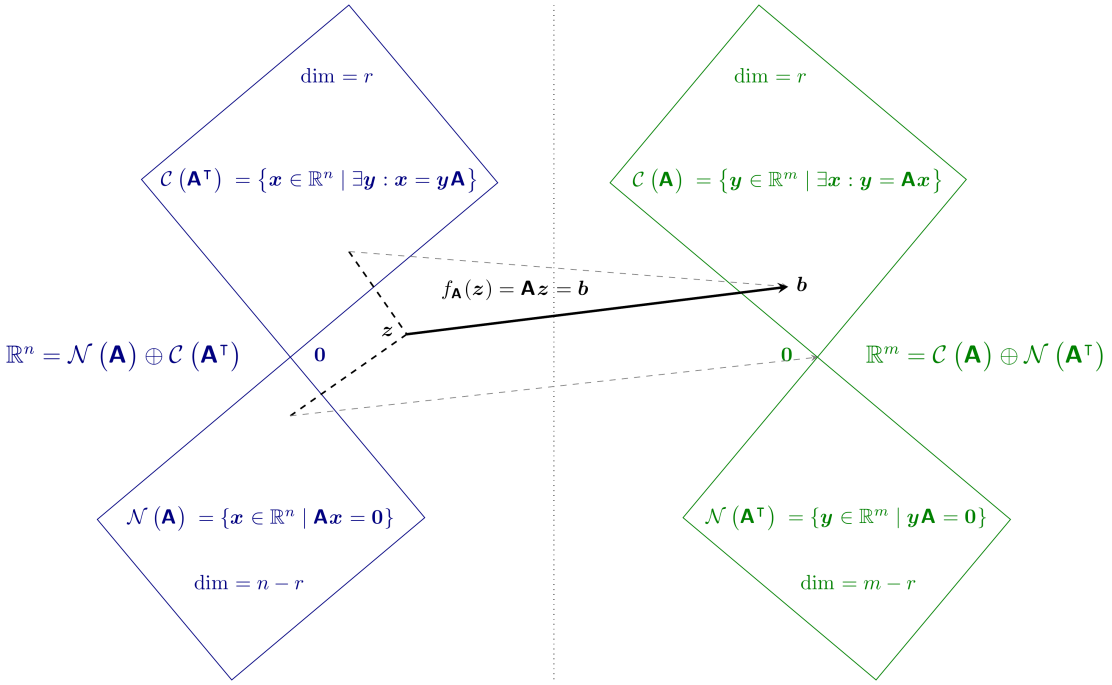
$$\boldsymbol{x} \in \mathcal{N}\left(\mathbf{A}\right) \quad \Rightarrow \quad \boldsymbol{d}\mathbf{A}\boldsymbol{x} = \boldsymbol{d} \cdot \mathbf{0} = 0.$$

null space $\perp$ row space	$\mathcal{N}\left(\mathbf{A}\right) \perp \mathcal{C}\left(\mathbf{A}^\intercal\right)$

Also: $\boldsymbol{x}\mathbf{A} = \mathbf{0} \quad \Rightarrow \quad \mathcal{N}\left(\mathbf{A}^\intercal\right) \perp \mathcal{C}\left(\mathbf{A}\right)$

F6

(**Lecture** 11) S-7 **The big picture: direct sum of orthogonal complements**



$\mathcal{C}\left(\mathbf{A}^\intercal\right) \perp \mathcal{N}\left(\mathbf{A}\right)$
$\boldsymbol{f} \cdot \boldsymbol{x} = \boldsymbol{y}\mathbf{A}\boldsymbol{x} = \boldsymbol{y} \cdot \mathbf{0}$

$\mathcal{C}\left(\mathbf{A}\right) \perp \mathcal{N}\left(\mathbf{A}^\intercal\right)$
$\boldsymbol{y} \cdot \boldsymbol{b} = \boldsymbol{y}\mathbf{A}\boldsymbol{x} = \mathbf{0} \cdot \boldsymbol{x}$

F7

(**Lecture** 11) S-8 **Revisiting Gaussian elimination**

An algorithm to find a basis for the orthogonal complement

Give me some vectors (I write them as rows of $\mathbf{M}$) and ...

$$\begin{bmatrix}\mathbf{M}\\ \mathbf{I}\end{bmatrix} = \left[\begin{array}{cccc} 1 & -3 & 0 & -1 \\ 0 & -1 & 1 & 1 \\ 1 & -4 & 1 & 0 \\ \hline 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[(3)\mathbf{1}+\mathbf{2}]\\ [(1)\mathbf{1}+\mathbf{4}]\\ [(1)\mathbf{2}+\mathbf{3}]\\ [(1)\mathbf{2}+\mathbf{4}]}]{\tau} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ \hline 1 & 3 & 3 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] = \begin{bmatrix}\mathbf{L}\\ \mathbf{E}\end{bmatrix} = \left[\begin{array}{cc} \mathbf{C} & \mathbf{0} \\ \hline \mathbf{D} & \mathbf{N} \end{array}\right]$$

Basis for the span of the given (row) vectors: $\mathcal{V}$

Basis for the orthogonal complement: $\mathcal{V}^{\perp}$ $\mathbf{MN} = \mathbf{0}$

But if you had given me $\mathbf{N}_{|1}$ and $\mathbf{N}_{|2}$, and I start anew... I would obtain a basis for... F8

The lecture ends here

Questions of the Lecture 11

(L-11) Question 1. Describe the set of vectors in $\mathbb{R}^3$ orthogonal to $\begin{pmatrix}1\\3\\-1\end{pmatrix}$

(Hefferon, 2008, exercise 2.15 from section II.2.)

(L-11) Question 2. Is there any vector that is perpendicular to itself?
(Hefferon, 2008, exercise 2.17 from section II.2.)

(L-11) Question 3. Find the length of each vector

(a) $\begin{pmatrix}1\\3\end{pmatrix}$. (b) $\begin{pmatrix}-1\\2\end{pmatrix}$. (c) $\begin{pmatrix}4\\1\\1\end{pmatrix}$.

(d) $\begin{pmatrix}0\\0\\0\end{pmatrix}$. (e) $\begin{pmatrix}1\\-1\\1\\0\end{pmatrix}$.

(Hefferon, 2008, exercise 2.11 from section II.2.)

(L-11) Question 4. Find a unit vector with the same direction as $\boldsymbol{v} = (2, -1, 0, 4, -2)$.

(L-11) Question 5. Find k so that these two vectors are perpendicular.

$$\begin{pmatrix} k, & 1, \end{pmatrix}, \qquad \begin{pmatrix} 4, & 3, \end{pmatrix}.$$

(Hefferon, 2008, exercise 2.14 from section II.2.)

(L-11) Question 6. Construct a matrix with the required property or state why that is impossible:

(a) Column space contains $\begin{pmatrix}1\\2\\-3\end{pmatrix}$ and $\begin{pmatrix}2\\-3\\5\end{pmatrix}$, the null space contains $\begin{pmatrix}1\\1\\1\end{pmatrix}$

(b) Row space contains $\begin{pmatrix}1\\2\\-3\end{pmatrix}$ and $\begin{pmatrix}2\\-3\\5\end{pmatrix}$, and the null space contains $\begin{pmatrix}1\\1\\1\end{pmatrix}$

(c) $\mathbf{A}\boldsymbol{x} = \begin{pmatrix}1\\1\\1\end{pmatrix}$ has a solution and $\mathbf{A}^{\mathsf{T}}\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$

(d) Each row is orthogonal to every column (and $\mathbf{A}$ is not the zero matrix)
(e) Columns add up to a column of zeros; rows add up to a row of 1's.

(Strang, 2003, exercise 3 from section 4.1.)

(L-11) Question 7. If $\mathbf{AB} = \mathbf{0}$, the columns of $\mathbf{B}$ are in the _____ of $\mathbf{A}$. The rows of $\mathbf{A}$ are in the _____ of $\mathbf{B}$. Why can't $\mathbf{A}$ and $\mathbf{B}$ be 3 by 3 matrices of rank 2?
(Strang, 2003, exercise 4 from section 4.1.)

(L-11) Question 8. Suppose that $\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{u} \cdot \boldsymbol{w}$ and $\boldsymbol{u} \neq \mathbf{0}$. Must $\boldsymbol{v} = \boldsymbol{w}$?
(Hefferon, 2008, exercise 2.20 from section II.2.)

(L-11) Question 9.

(a) If $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ has a solution and $\mathbf{A}^\intercal \boldsymbol{y} = \mathbf{0}$, then $\boldsymbol{y}$ is perpendicular to ____.
(b) If $\mathbf{A}^\intercal \boldsymbol{y} = \boldsymbol{c}$ has a solution and $\mathbf{A}\boldsymbol{x} = \mathbf{0}$, then $\boldsymbol{x}$ is perpendicular to ____.

(Strang, 2003, exercise 5 from section 4.1.)

(L-11) Question 10. Demonstrate, for $\mathbb{R}^n$, that if $\boldsymbol{u}$ and $\boldsymbol{v}$ are perpendicular then $||\boldsymbol{u} + \boldsymbol{v}||^2 = ||\boldsymbol{u}||^2 + ||\boldsymbol{v}||^2$.
(Hefferon, 2008, exercise 2.33 from section II.2.)

(L-11) Question 11.

(a) Find a 1 by 3 matrix whose null space consists of all vectors in $\mathbb{R}^3$ such that $x_1 + 2x_2 + 4x_3 = 0$. Find a 3 by 3 matrix with that same null space.
(b) Find a 1 by 3 matrix whose null space is orthogonal to all vectors in $\mathbb{R}^3$ such that $x_1 + 2x_2 + 4x_3 = 0$. Find a 3 by 3 matrix with that same null space.

(Strang, 2006, exercise 9 from section 2.4.)

(L-11) Question 12. Consider $\mathbf{A}$ with exactly two special solutions for $\boldsymbol{x}\mathbf{A} = \mathbf{0}$:

$\boldsymbol{s}_1 = \begin{pmatrix} 3, & 1, & 0, & 0, \end{pmatrix}$, and $\underset{4\times 2}{\boldsymbol{s}_2} = \begin{pmatrix} 6, & 0, & 2, & 1, \end{pmatrix}$.

(a) Find the reduced row echelon form $\mathbf{R}$ of $\mathbf{A}$.
(b) What is the row space of $\mathbf{A}$?
(c) What is the complete solution to $\boldsymbol{x}\mathbf{R} = \begin{pmatrix} 3, & 6, \end{pmatrix}$?
(d) Find a combination of rows 2, 3, 4 that equals $\mathbf{0}$. (Not OK to use $0({}_{2|}\mathbf{A}) + 0({}_{3|}\mathbf{A}) + 0({}_{4|}\mathbf{A})$. The problem is to show that these rows are dependent.)

basado en MIT Course 18.06 Quiz 1, March 4, 2013

(L-11) Question 13. Suppose $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ has a solution (maybe many solutions). It can be shown that any solution $\boldsymbol{x}$ of this system can be decomposed as the sum of two vectors ($\boldsymbol{x} = \boldsymbol{x}_r + \boldsymbol{x}_n$) where $\boldsymbol{x}_r$ is a combination of the rows of $\mathbf{A}$ and $\boldsymbol{x}_n$ belongs to the solution set of $\mathbf{A}\boldsymbol{x} = \mathbf{0}$.

(a) (0.5^{pts}) Prove that $\mathbf{A}(\boldsymbol{x}_r) = \boldsymbol{b}$.
(b) (1^{pts}) Suppose that $\boldsymbol{v}_r$ is a linear combination of the rows of $\mathbf{A}$ and furthermore $\mathbf{A}(\boldsymbol{v}_r) = \boldsymbol{b}$. What vector subspaces does the difference $(\boldsymbol{v}_r - \boldsymbol{x}_r)$ belong to? Show that $\boldsymbol{x}_r$ and $\boldsymbol{v}_r$ are equal.
(c) (1^{pts}) Compute the solution $\boldsymbol{x}_r$ in the row space of this matrix $\mathbf{A}$, by solving for c and d

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \end{bmatrix} \boldsymbol{x}_r = \begin{pmatrix} 14 \\ 9 \end{pmatrix} \quad \text{with} \quad \boldsymbol{x}_r = c \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + d \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}.$$

End of Questions of the Lecture 11

LECTURE 12: Parametric and Cartesian equations

Lecture 12

(**Lecture** 12) S-1 Highlights of Lecture 12

Highlights of *Lecture 12*

- From parametric to Cartesian (or implicit) equations
- Choosing among parametric equations

F9

De ecuaciones paramétricas a ecuaciones cartesianas

(**Lecture** 12) S-2 Cartesian (implicit) and parametric equations of lines and planes

Cartesian (implicit) equations $\{\boldsymbol{x} \in \mathbb{R}^n \mid \mathbf{A}\boldsymbol{x} = \boldsymbol{b}\}$**:**
For example

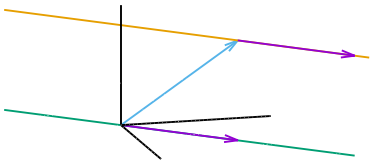
$$\left\{\boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \boldsymbol{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\} = \text{sol. set of } \begin{cases} x_1 - x_2 + x_3 = 1 \\ x_3 = 1 \end{cases}$$

Parametric equations:
for the above set

$$\left\{\boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^1 : \boldsymbol{x} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \boldsymbol{p}\right\}$$

In this case *dimension* 1 A line (there is only one parameter a)
line line

F10

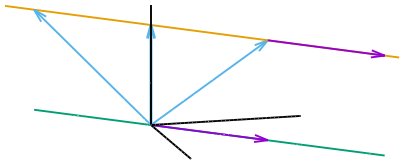


or

$$\left\{\boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^1 : \boldsymbol{x} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \boldsymbol{p}\right\}$$

or

$$\left\{ \boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^1 : \boldsymbol{x} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \boldsymbol{p} \right\}$$



(**Lecture 12**) S-3 **Cartesian (implicit) and parametric equations of lines and planes**

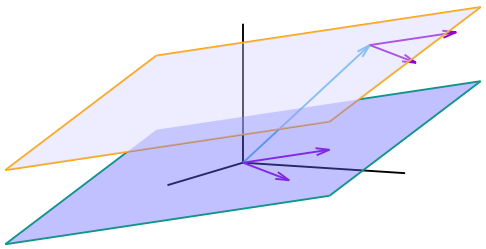
Cartesian (implicit) equations $\{\boldsymbol{x} \in \mathbb{R}^n \mid \mathbf{A}\boldsymbol{x} = \boldsymbol{b}\}$**:**
For example

$$\left\{ \boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \boldsymbol{x} = \begin{pmatrix} 1, \end{pmatrix} \right\} = \text{sol. set of } \{x_1 - x_2 + x_3 = 1$$

Parametric equations:
for the above set

$$\left\{ \boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^2 : \boldsymbol{x} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \boldsymbol{p} \right\}$$

In this case *dimension* 2 A plane (there are two parameters a and b)
plane plane
F12

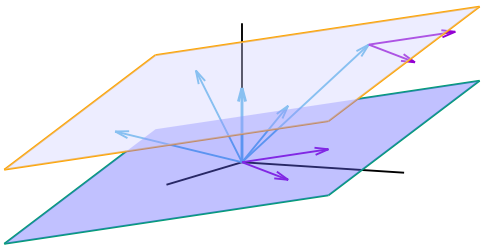


or

$$\left\{ \boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^2 : \boldsymbol{x} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \boldsymbol{p} \right\}$$

but also

$$\left\{ \boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^2 : \boldsymbol{x} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} + \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \boldsymbol{p} \right\}$$



(**Lecture 12**) **S-4** **From parametric to Cartesian equations**

$$\boxed{\mathcal{C}\left(\mathbf{A}^{\intercal}\right) \perp \mathcal{N}\left(\mathbf{A}\right)}$$

Consider

$$H = \left\{ \boldsymbol{x} \in \mathbb{R}^n \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^k : \boldsymbol{x} = \boldsymbol{s} + \left[\boldsymbol{n}_1;\ \ldots\ \boldsymbol{n}_k;\right]\boldsymbol{p} \right\}.$$

If we find $\mathbf{A}$ *such that* $\mathbf{A}\boldsymbol{n}_i = \mathbf{0}$, then if $\boldsymbol{x} \in H$

$$\mathbf{A}\boldsymbol{x} \;=\; \mathbf{A}\boldsymbol{s} \;+\; \underbrace{\mathbf{A}\left[\boldsymbol{n}_1;\ \ldots\ \boldsymbol{n}_k;\right]}_{\mathbf{0}}\boldsymbol{p} \quad \Rightarrow \quad \mathbf{A}\boldsymbol{x} \;=\; \boldsymbol{b}, \quad \text{where } \boldsymbol{b} = \mathbf{A}\boldsymbol{s}.$$

Therefore

$$H = \left\{ \boldsymbol{x} \in \mathbb{R}^n \mid \mathbf{A}\boldsymbol{x} = \boldsymbol{b} \right\}.$$

F14

(**Lecture 12**) **S-5** **From the set of solutions to a linear system**

Find the implicit equations of the plane P parallel to the span of $(1,2,0,-2)$ and $(0,0,1,3)$, that goes through $\boldsymbol{s} = (1,3,1,1)$.

$$P = \left\{ \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} \;\middle|\; \exists a, b \in \mathbb{R} : \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 1 \\ 1 \end{pmatrix} + a \begin{pmatrix} 1 \\ 2 \\ 0 \\ -2 \end{pmatrix} + b \begin{pmatrix} 0 \\ 0 \\ 1 \\ 3 \end{pmatrix} \right\}$$

$$= \left\{ \boldsymbol{x} \in \mathbb{R}^4 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^2 : \boldsymbol{x} = \begin{pmatrix} 1 \\ 3 \\ 1 \\ 1 \end{pmatrix} + \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 1 \\ -2 & 3 \end{bmatrix} \boldsymbol{p} \right\}$$

We need vectors perpendicular to $(1,2,0,-2)$ and $(0,0,1,3)$

F15

(**Lecture 12**) **S-6** **From the set of solutions to a linear system**

$\boldsymbol{x} = (x, y, z, w,)$; $\boldsymbol{s} = (1, 3, 1, 1,)$.

$$\left[\begin{array}{cccc} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 3 \\ \hline x & y & z & w \\ \hline 1 & 3 & 1 & 1 \end{array}\right] \xrightarrow[[(2)\mathbf{1}+\mathbf{4}]]{\boldsymbol{\tau}[(-2)\mathbf{1}+\mathbf{2}]} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ \hline x & y-2x & z & w+2x \\ \hline 1 & 1 & 1 & 3 \end{array}\right] \xrightarrow{\boldsymbol{\tau}[(-3)\mathbf{3}+\mathbf{4}]} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline x & y-2x & z & w+2x-3z \\ \hline 1 & 1 & 1 & 0 \end{array}\right]$$

So $\mathbf{A} = \begin{bmatrix} -2 & 1 & 0 & 0 \\ 2 & 0 & -3 & 1 \end{bmatrix}$; and then $\mathbf{A}\boldsymbol{x} = \begin{pmatrix} -2x + y \\ 2x + w - 3z \end{pmatrix}$ and $\boldsymbol{b} = \mathbf{A}\boldsymbol{s} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Hence $\begin{cases} -2x + y = 1 \\ 2x - 3z + w = 0 \end{cases}$

$$P = \left\{ \boldsymbol{x} \in \mathbb{R}^4 \;\middle|\; \begin{bmatrix} -2 & 1 & 0 & 0 \\ 2 & 0 & -3 & 1 \end{bmatrix} \boldsymbol{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}.$$

F16

De unas ecuaciones paramétricas a otras

Cuando resolvemos un sistema de ecuaciones de un problema aplicado, puede ocurrir que estemos interesados en alguna ecuación paramétrica en especial (es lo que comúnmente se llama despejar unas variables respecto de las otras). Veamos cómo hacerlo una vez hemos resuelto un sistema, es decir, veamos cómo pasar de unas ecuaciones paramétricas a otras.

(**Lecture 12**) **S-7** **A problem from Microeconomics**

Solve for Y in terms of X to obtain the PPF

$$\begin{cases} X = 4L_x \\ Y = 3L_y \\ L_x + L_y = 80 \end{cases} \rightarrow \begin{cases} X - 4L_x = 0 \\ Y - 3L_y = 0 \\ L_x + L_y = 80 \end{cases}$$

(*"in terms of" X means X free*)

$$\left[\begin{array}{cccc|c} 1 & 0 & -4 & 0 & 0 \\ 0 & 1 & 0 & -3 & 0 \\ 0 & 0 & 1 & 1 & -80 \\ \hline 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[[(3)\mathbf{2}+\mathbf{4}]]{\boldsymbol{\tau}[(4)\mathbf{1}+\mathbf{3}]} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & -80 \\ \hline 1 & 0 & 4 & 0 & 0 \\ 0 & 1 & 0 & 3 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\boldsymbol{\tau}[(80)\mathbf{3}+\mathbf{5}]} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ \hline 1 & 0 & 4 & -4 & 320 \\ 0 & 1 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 80 \\ 0 & 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\begin{pmatrix} X \\ Y \\ L_x \\ L_y \end{pmatrix} = \begin{pmatrix} 320 \\ 0 \\ 80 \\ 0 \end{pmatrix} + a \begin{pmatrix} -4 \\ 3 \\ -1 \\ 1 \end{pmatrix} \Rightarrow a = L_y \Rightarrow \begin{pmatrix} X \\ Y \\ L_x \\ L_y \end{pmatrix} = \begin{pmatrix} 320 - 4L_y \\ 3L_y \\ 80 - L_y \\ L_y \end{pmatrix} \quad \text{"in terms of" } L_y$$

F17

(**Lecture 12**) **S-8** **Free variable**

$$\left[\begin{array}{cccc|c} 1&0&0&0&0\\ 0&1&0&0&0\\ 0&0&1&0&0\\ \hline 1&0&4&-4&320\\ 0&1&0&3&0\\ 0&0&1&-1&80\\ 0&0&0&1&0\\ \hline 0&0&0&0&1 \end{array}\right] \xrightarrow[\text{[(-320)4+5]}]{\tau\ [(\frac{-1}{4})4]} \left[\begin{array}{cccc|c} 1&0&0&0&0\\ 0&1&0&0&0\\ 0&0&1&0&0\\ \hline 1&0&4&1&0\\ 0&1&0&-3/4&240\\ 0&0&1&1/4&0\\ 0&0&0&-1/4&80\\ \hline 0&0&0&0&1 \end{array}\right]$$

$$\begin{pmatrix} X\\ Y\\ L_x\\ L_y \end{pmatrix} = \begin{pmatrix} 0\\ 240\\ 0\\ 80 \end{pmatrix} + a \begin{pmatrix} 1\\ -\frac{3}{4}\\ \frac{1}{4}\\ -\frac{1}{4} \end{pmatrix} \Rightarrow\ a = X \quad \Rightarrow \quad \begin{pmatrix} X\\ Y\\ L_x\\ L_y \end{pmatrix} = \begin{pmatrix} X\\ 240-\frac{3}{4}X\\ \frac{1}{4}X\\ 80-\frac{1}{4}X \end{pmatrix}$$

"in terms of" X

F18

(**Lecture 12**) **S-9** **Free variables**

$$\begin{cases} x+2y- \ z+ \ w=1\\ -x-2y+3z+5w=5\\ -x-2y- \ z-7w=-7 \end{cases}$$

1. Solve in terms of y and w
2. Solve in terms of x and w
3. Solve in terms of x and z
4. Solve in terms of x and y

F19

$$\left[\begin{array}{cccc|c} 1&2&-1&1&-1\\ -1&-2&3&5&-5\\ -1&-2&-1&-7&7\\ \hline 1&0&0&0&0\\ 0&1&0&0&0\\ 0&0&1&0&0\\ 0&0&0&1&0\\ \hline 0&0&0&0&1 \end{array}\right] \xrightarrow[\substack{[(1)1+3]\\ [(-1)1+4]\\ [(1)1+5]}]{\tau\ [(-2)1+2]} \left[\begin{array}{cccc|c} 1&0&0&0&0\\ -1&0&2&6&-6\\ -1&0&-2&-6&6\\ \hline 1&-2&1&-1&1\\ 0&1&0&0&0\\ 0&0&1&0&0\\ 0&0&0&1&0\\ \hline 0&0&0&0&1 \end{array}\right] \xrightarrow[[(3)3+5]]{\tau\ [(-3)3+4]} \left[\begin{array}{cccc|c} 1&0&0&0&0\\ -1&0&2&0&0\\ -1&0&-2&0&0\\ \hline 1&-2&1&-4&4\\ 0&1&0&0&0\\ 0&0&1&-3&3\\ 0&0&0&1&0\\ \hline 0&0&0&0&1 \end{array}\right]$$

$$\left[\begin{array}{cc|c} -2&-4&4\\ 1&0&0\\ 0&-3&3\\ 0&1&0 \end{array}\right] \begin{cases} \xrightarrow[\substack{[(4)1+2]\\ [(-4)1+3]}]{\tau\ [(\frac{-1}{2})1]} \left[\begin{array}{cc|c} 1&0&0\\ \frac{-1}{2}&-2&2\\ 0&-3&3\\ 0&1&0 \end{array}\right] \xrightarrow[[(\frac{-1}{3})2]]{\tau\ [(1)2+3]} \left[\begin{array}{cc|c} 1&0&0\\ \frac{-1}{2}&\frac{2}{3}&0\\ 0&1&0\\ 0&\frac{-1}{3}&1 \end{array}\right]\\ \xrightarrow[\substack{[(4)1+2]\\ [(-4)1+3]}]{\tau\ [(\frac{-1}{2})1]} \left[\begin{array}{cc|c} 1&0&0\\ \frac{-1}{2}&-2&2\\ 0&-3&3\\ 0&1&0 \end{array}\right] \xrightarrow[[(\frac{1}{2})2+1]\\ [(-2)2+3]]{\tau\ [(\frac{-1}{2})2]} \left[\begin{array}{cc|c} 1&0&0\\ 0&1&0\\ \frac{3}{4}&\frac{3}{2}&0\\ \frac{-1}{4}&\frac{-1}{2}&1 \end{array}\right] \end{cases}$$

The lecture ends here

Questions of the Lecture 12

(L-12) Question 1.

(a) Find a parametric representation of the line passing through the points $\boldsymbol{x}_P = \begin{pmatrix}1\\2\end{pmatrix}$ and $\boldsymbol{x}_Q = \begin{pmatrix}3\\1\end{pmatrix}$.

(b) Find an implicit representation of the same line.

(L-12) Question 2.

(a) Find a parametric representation of the line passing through the points $\boldsymbol{x}_P = (1, \quad -3, \quad 1,)$ and $\boldsymbol{x}_Q = (-2, \quad 4, \quad 5,)$.

(b) Find an implicit representation (Cartesian equations) for the same line.

(Lang, 1986, Example 1 in Section 1.5)

(L-12) Question 3.

(a) Parametric equation of a line parallel to $2x - 3y = 5$ that goes through $(1, 1)$.
(b) Find a implicit representation for the line.

(L-12) Question 4.

(a) Find the parametric equations of the plane that passes through the point $(0, \quad 1, \quad 1,)$ and is parallel to the vectors $(0, \quad 1, \quad 2,)$ and $(1, \quad 1, \quad 0,)$.
(b) Write the implicit equation of the same plane.

(L-12) Question 5.

(a) Find a parametric equation of the plane that passes through the point $(2, \quad 1, \quad 3,)$ and is perpendicular to the vector $(3, \quad 1, \quad 1,)$.
(b) Write the implicit equation of the same plane.

(L-12) Question 6. Consider the system of equations $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 & 1 & 1 \\ 0 & 0 & 2 & 3 & 1 \\ 0 & 0 & 1 & 4 & 2 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix}.$$

(a) (1^{pts}) Find the solution to the system.
(b) (0.5^{pts}) Explain why the solution set to the previous system is a line in $\mathbb{R}^5$. Find a direction vector (a vector parallel to the line) and any point on that line.
(c) (1^{pts}) Find the set of vectors perpendicular to the previous direction vector. Prove that this set of perpendicular vectors forms a four-dimensional vector subspace. Find a basis for this subspace.

End of Questions of the Lecture 12

LECTURE 13: Projections onto Subspaces

Lecture 13

(**Lecture** 13) S-1 Outline of Lesson 13

Outline of *Lesson 13*

- Projections
- Projection Matrices

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(**Lecture** 13) S-2 Direct Sum of Subspaces

$\mathbb{R}^n$ is a *direct sum* of $\mathcal{A}$ and $\mathcal{B}$ $\quad (\mathbb{R}^n = \mathcal{A} \oplus \mathcal{B})$

if every $\boldsymbol{x} \in \mathbb{R}^n$ has a **unique** representation $\boldsymbol{x} = \boldsymbol{a} + \boldsymbol{b}$,

with $\boldsymbol{a} \in \mathcal{A}$ and $\boldsymbol{b} \in \mathcal{B}$.

Example 1. $\boxed{\mathbb{R}^n = \mathcal{C}\left(\mathbf{A}^{\mathsf{T}}\right) \oplus \mathcal{N}\left(\mathbf{A}\right)}$

$$\begin{bmatrix} \mathbf{A} \\ \mathbf{I} \end{bmatrix} = \left[\begin{array}{ccc} 1 & 2 & 5 \\ 2 & 4 & 10 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc} 1 & 0 & 0 \\ 2 & 0 & 0 \\ \hline 1 & -2 & -5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \Rightarrow \text{Basis of } \mathbb{R}^3; \left[\begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}; \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}; \begin{pmatrix} -5 \\ 0 \\ 1 \end{pmatrix}\right]$$

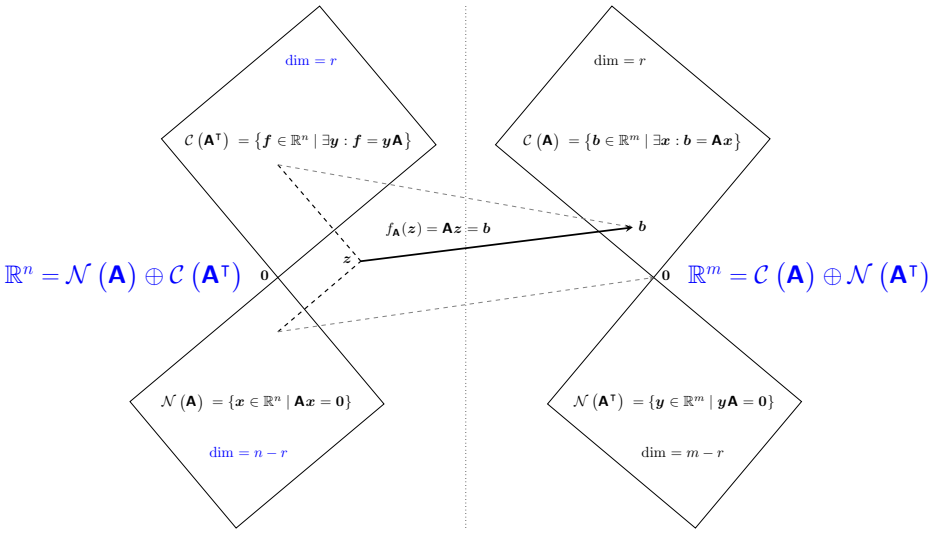
$$\forall \boldsymbol{x} \in \mathbb{R}^3,\ \exists c_1, c_2, c_3 \ \Big|\ \boldsymbol{x} = c_1 \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} + c_2 \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} -5 \\ 0 \\ 1 \end{pmatrix} = \boldsymbol{a} + \boldsymbol{b}$$

where $\boldsymbol{a} \in \mathcal{C}\left(\mathbf{A}^{\mathsf{T}}\right)$ and $\boldsymbol{b} \in \mathcal{N}\left(\mathbf{A}\right)$.

Also $\boxed{\mathbb{R}^m = \mathcal{C}\left(\mathbf{A}\right) \oplus \mathcal{N}\left(\mathbf{A}^{\mathsf{T}}\right)}$

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(**Lecture 13**) **S-3** **The Big Picture: Direct Sum of Orthogonal Complements**



$$\mathcal{C}(\mathbf{A}^\intercal) \perp \mathcal{N}(\mathbf{A}) \qquad \mathcal{C}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A}^\intercal)$$

$$\boldsymbol{f} \cdot \boldsymbol{x} = \boldsymbol{y}\mathbf{A}\boldsymbol{x} = \boldsymbol{y} \cdot \mathbf{0} \qquad \boldsymbol{y} \cdot \boldsymbol{b} = \boldsymbol{y}\mathbf{A}\boldsymbol{x} = \mathbf{0} \cdot \boldsymbol{x}$$

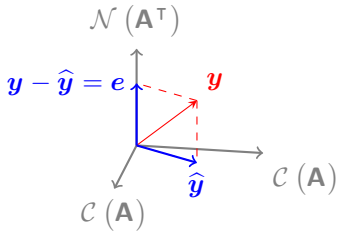
F24

(**Lecture 13**) **S-4** **Orthogonal Projection onto $\mathcal{C}(\mathbf{A})$**

Consider $\underset{m\times n}{\mathbf{A}}$; since $\mathbb{R}^m = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^\intercal)$, for any $\boldsymbol{y} \in \mathbb{R}^m$

$$\boldsymbol{y} = \widehat{\boldsymbol{y}} + \boldsymbol{e}; \qquad (\boldsymbol{e} = \boldsymbol{y} - \widehat{\boldsymbol{y}})$$

where $\boxed{\widehat{\boldsymbol{y}} \in \mathcal{C}(\mathbf{A}) \text{ and } \boldsymbol{e} \perp \widehat{\boldsymbol{y}}}$, thus $\boldsymbol{e} \in \mathcal{N}(\mathbf{A}^\intercal)$.



How to compute $\widehat{\boldsymbol{y}} \in \mathcal{C}(\mathbf{A})$?

F25

(Lecture 13) S-5 Normal Equations

Consider $\underset{m\times n}{\mathbf{A}}$. We want to find the decomposition $\boxed{\boldsymbol{y}=\widehat{\boldsymbol{y}}+\boldsymbol{e}}$ where

$$\widehat{\boldsymbol{y}}\in\mathcal{C}\left(\mathbf{A}\right) \qquad \text{and} \qquad (\widehat{\boldsymbol{y}}-\boldsymbol{y})\in\mathcal{N}\left(\mathbf{A}^{\intercal}\right)$$

Thus

$$\mathbf{A}\widehat{\boldsymbol{x}}=\widehat{\boldsymbol{y}} \qquad \Leftrightarrow \qquad (\mathbf{A}\widehat{\boldsymbol{x}}-\boldsymbol{y})\in\mathcal{N}\left(\mathbf{A}^{\intercal}\right)$$

That is

$$\mathbf{A}\widehat{\boldsymbol{x}}=\widehat{\boldsymbol{y}} \quad \Leftrightarrow \quad \mathbf{A}^{\intercal}\left(\mathbf{A}\widehat{\boldsymbol{x}}-\boldsymbol{y}\right)=\mathbf{0} \quad \Leftrightarrow \quad \boxed{(\mathbf{A}^{\intercal}\mathbf{A})\widehat{\boldsymbol{x}}=\mathbf{A}^{\intercal}\boldsymbol{y}}$$

Equivalent systems! $\Rightarrow \mathcal{N}\left(\mathbf{A}\right)=\mathcal{N}\left(\mathbf{A}^{\intercal}\mathbf{A}\right) \Rightarrow \operatorname{rg}(\mathbf{A})=\operatorname{rg}(\mathbf{A}^{\intercal}\mathbf{A})$

unique solution $\widehat{\boldsymbol{x}}$ if and only if $\mathbf{A}$ has independent columns

F26

(Lecture 13) S-6 Solution to the Normal Equations (Full Column Rank)

$\boxed{\mathbf{A}^{\intercal}\mathbf{A}\widehat{\boldsymbol{x}}=\mathbf{A}^{\intercal}\boldsymbol{y}}$ ($\mathbf{A}$ is full column rank)

The Solution $\widehat{\boldsymbol{x}}=(\mathbf{A}^{\intercal}\mathbf{A})^{-1}\mathbf{A}^{\intercal}\boldsymbol{y}$

The Projection $\widehat{\boldsymbol{y}}=\mathbf{A}\widehat{\boldsymbol{x}}=\mathbf{A}(\mathbf{A}^{\intercal}\mathbf{A})^{-1}\mathbf{A}^{\intercal}\boldsymbol{y}$

The Projection Matrix $\mathbf{P}=\mathbf{A}(\mathbf{A}^{\intercal}\mathbf{A})^{-1}\mathbf{A}^{\intercal}$

$$\widehat{\boldsymbol{y}}=\mathbf{P}\boldsymbol{y}$$

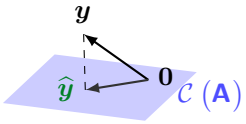
$\mathbf{P}$: Symmetric and Idempotent.

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(Lecture 13) S-7 Projection Matrix

$$\boxed{\mathbf{P}=\mathbf{A}\left(\mathbf{A}^{\intercal}\mathbf{A}\right)^{-1}\mathbf{A}^{\intercal}}$$

The Projection $\mathbf{P}\boldsymbol{y}$ *is the point* $\widehat{\boldsymbol{y}}$ *closest to* $\boldsymbol{y}$

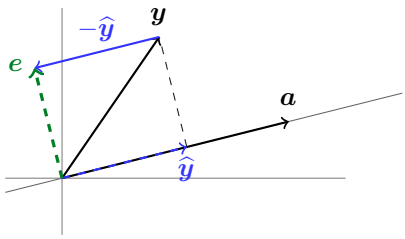


Extreme Cases:

- If $\boldsymbol{y}\in\mathcal{C}\left(\mathbf{A}\right)$ then $\mathbf{P}\boldsymbol{y}=$
- If $\boldsymbol{y}\perp\mathcal{C}\left(\mathbf{A}\right)$ then $\mathbf{P}\boldsymbol{y}=$

F28

(Lecture 13) **S-8 Projection onto a Line**



line$= \mathcal{C}(\mathbf{A})$; $\mathbf{A} = [a]$; $a \neq \mathbf{0}$

We want to find the point $\widehat{y}$ on the line that is closest to y
$\widehat{y} \in \mathcal{C}([a])$ $\perp$ $e = (y - \widehat{y}) \in \mathcal{N}([a]^{\mathsf{T}})$.

$\widehat{y}$ is some multiple of a:

$$\widehat{y} = [a](\widehat{x},)$$

How:

$$[a]^{\mathsf{T}}[a]\widehat{x} = [a]^{\mathsf{T}}y$$

The Solution

$$\widehat{x} = ([a]^{\mathsf{T}}[a])^{-1}[a]^{\mathsf{T}}y$$

The Projection

$$\widehat{y} = [a]\widehat{x} = [a]([a]^{\mathsf{T}}[a])^{-1}[a]^{\mathsf{T}}y$$

The Projection Matrix

$$\mathbf{P} = [a]([a]^{\mathsf{T}}[a])^{-1}[a]^{\mathsf{T}}$$

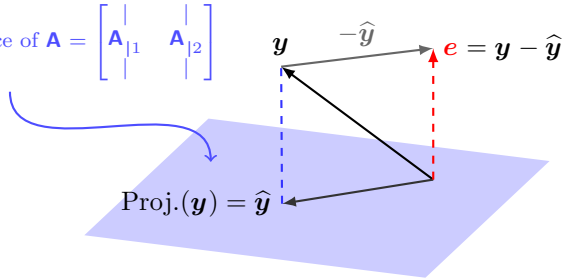
F29

(Lecture 13) **S-9 Projection onto a plane**

Why project?
So we will solve

$$\mathbf{A}x = \Big(\text{Proj. of } y \text{ onto } \mathcal{C}(\mathbf{A})\Big).$$

$\mathcal{C}(\mathbf{A})$: Column Space of $\mathbf{A} = \begin{bmatrix} | & | \\ \mathbf{A}_{|1} & \mathbf{A}_{|2} \\ | & | \end{bmatrix}$

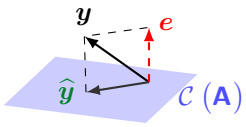


$(y - \widehat{y}) = e \perp \mathcal{C}(\mathbf{A})$ ...that's the fundamental fact.

F30

(Lecture 13) **S-10** Normal Equations

What's the projection of $\boldsymbol{y}$ onto the column space of $\mathbf{A}=\begin{bmatrix} | & | \\ \mathbf{A}_{|1} & \mathbf{A}_{|2} \\ | & | \end{bmatrix}$?



$$\widehat{\boldsymbol{y}} = (\mathbf{A}_{|1})\widehat{x_1} + (\mathbf{A}_{|2})\widehat{x_2} \;=\; \mathbf{A}\widehat{\boldsymbol{x}}$$

"Find a combination of the columns such that $\boldsymbol{e} \perp \mathcal{C}\,(\mathbf{A})$"

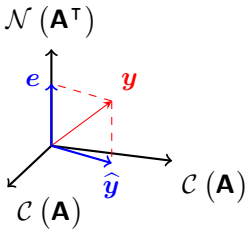
$$\boldsymbol{e} \perp \mathcal{C}\,(\mathbf{A}) \quad \Rightarrow \quad \boldsymbol{e} \in$$

$$\mathbf{A}^{\mathsf{T}}\boldsymbol{e} = \mathbf{A}^{\mathsf{T}}(\boldsymbol{y} - \widehat{\boldsymbol{y}}) \quad = \quad \mathbf{A}^{\mathsf{T}}(\boldsymbol{y} - \mathbf{A}\widehat{\boldsymbol{x}}) = \mathbf{0} \quad \Leftrightarrow \quad \boxed{(\mathbf{A}^{\mathsf{T}}\mathbf{A})\widehat{\boldsymbol{x}} = \mathbf{A}^{\mathsf{T}}\boldsymbol{y}}$$

F31

(Lecture 13) **S-11** Two Projections

$\boldsymbol{y}$ has a component $\widehat{\boldsymbol{y}}$ in $\mathcal{C}\,(\mathbf{A})$, and another component $\boldsymbol{e}$ in $\mathcal{C}\,(\mathbf{A})^{\perp}$.



$$\widehat{\boldsymbol{y}} + \boldsymbol{e} = \boldsymbol{y}$$

$\widehat{\boldsymbol{y}} = \mathbf{P}\boldsymbol{y}$ is the projection onto $\mathcal{C}\,(\mathbf{A})$

$\boldsymbol{e} = (\mathbf{I} - \mathbf{P})\boldsymbol{y}$ projection onto $\mathcal{C}\,(\mathbf{A})^{\perp}$

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The lecture ends here

Questions of the Lecture 13

(L-13) Question 1. Project the first vector orthogonally onto the line spanned by the second vector. Verify that $\boldsymbol{e}$ is perpendicular to $\boldsymbol{a}$. Find the projection matrix $\mathbf{P} = [\boldsymbol{a}]\big([\boldsymbol{a}]^{\mathsf{T}}[\boldsymbol{a}]\big)^{-1}[\boldsymbol{a}]^{\mathsf{T}}$ onto the line through each vector $\boldsymbol{a}$. Verify in each case that $\mathbf{P}^2 = \mathbf{P}$. Multiply $\mathbf{P}\boldsymbol{b}$ in each case to compute the projection $\widehat{\boldsymbol{b}}$.

(a) $\boldsymbol{b} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$; $\boldsymbol{a} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$.

(b) $\boldsymbol{b} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$; $\boldsymbol{a} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$.

(c) $\boldsymbol{b} = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix}$; $\boldsymbol{a} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$.

(d) $\boldsymbol{b} = \begin{pmatrix}1\\1\\4\end{pmatrix}$; $\boldsymbol{a} = \begin{pmatrix}3\\3\\12\end{pmatrix}$.

(Hefferon, 2008, exercise 1.6 from section VI.1.)

(L-13) Question 2. Project the vector orthogonally onto the line.

(a) $\begin{pmatrix}2\\-1\\4\end{pmatrix}$, The line : $\left\{ \boldsymbol{v} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^1,\ \boldsymbol{v} = \begin{bmatrix}-3\\1\\-3\end{bmatrix} \boldsymbol{p} \right\}$.

(b) $\begin{pmatrix}-1\\-1\end{pmatrix}$, the line described by the equation $y = 3x$.

(L-13) Question 3. Although diagrams have guided our development of the topic, we are not restricted to spaces that we can draw. In $\mathbb{R}^4$ project the vector onto the line.

$$\begin{pmatrix}1\\2\\1\\3\end{pmatrix}; \quad \left\{ \boldsymbol{v} \in \mathbb{R}^4 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^1,\ \boldsymbol{v} = \begin{bmatrix}-1\\1\\-1\\1\end{bmatrix} \boldsymbol{p} \right\}.$$

(L-13) Question 4.

(a) Project the vector $\boldsymbol{b} = \begin{pmatrix}1, & 1,\end{pmatrix}$ onto the lines spanned by $\boldsymbol{a}_1 = \begin{pmatrix}1, & 0,\end{pmatrix}$ and $\boldsymbol{a}_2 = \begin{pmatrix}1, & 2,\end{pmatrix}$. Add the projections: $\widehat{\boldsymbol{b}_1} + \widehat{\boldsymbol{b}_2}$. The projections do not add to $\boldsymbol{b}$ because the vectors $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$ are not orthogonal.

(b) The projection of $\boldsymbol{b}$ onto the plane spanned by $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$ will equal $\boldsymbol{b}$. Find $\mathsf{P} = \mathsf{A}(\mathsf{A}^\mathsf{T}\mathsf{A})^{-1}\mathsf{A}^\mathsf{T}$ for $\mathsf{A} = \begin{bmatrix}\boldsymbol{a}_1; & \boldsymbol{a}_2;\end{bmatrix}$.

(Strang, 2003, exercise 8–9 from section 4.2.)

(L-13) Question 5.

(a) If $\mathsf{P}^2 = \mathsf{P}$ show that $(\mathsf{I} - \mathsf{P})^2 = \mathsf{I} - \mathsf{P}$. When P projects onto the column space of A, $(\mathsf{I} - \mathsf{P})$ projects onto the _____.

(b) If $\mathsf{P}^\mathsf{T} = \mathsf{P}$ show that $(\mathsf{I} - \mathsf{P})^\mathsf{T} = \mathsf{I} - \mathsf{P}$.

(Strang, 2003, exercise 17 from section 4.2.)

(L-13) Question 6.

(a) Compute the projection matrices $\mathsf{P} = \begin{bmatrix}\boldsymbol{a}\end{bmatrix}\left(\begin{bmatrix}\boldsymbol{a}\end{bmatrix}^\mathsf{T}\begin{bmatrix}\boldsymbol{a}\end{bmatrix}\right)^{-1}\begin{bmatrix}\boldsymbol{a}\end{bmatrix}^\mathsf{T}$ onto the lines through $\boldsymbol{a}_1 = \begin{pmatrix}-1, & 2, & 2,\end{pmatrix}$ and $\boldsymbol{a}_2 = \begin{pmatrix}2, & 2, & -1,\end{pmatrix}$. Show that $\boldsymbol{a}_1 \perp \boldsymbol{a}_2$. Multiply those projection matrices and explain why their product $\mathsf{P}_1\mathsf{P}_2$ is what it is.

(b) Project $\boldsymbol{b} = \begin{pmatrix}1, & 0, & 0,\end{pmatrix}$ onto the lines through $\boldsymbol{a}_1$, and $\boldsymbol{a}_2$ and also onto $\boldsymbol{a}_3 = \begin{pmatrix}2, & -1, & 2,\end{pmatrix}$. Add the three projections $\widehat{\boldsymbol{b}_1} + \widehat{\boldsymbol{b}_2} + \widehat{\boldsymbol{b}_3}$.

(c) Find the projection matrix P_3 onto $\mathcal{L}\left(\begin{bmatrix}\boldsymbol{a}_3;\end{bmatrix}\right) = \mathcal{L}\left(\begin{bmatrix}\begin{pmatrix}2, & -1, & 2,\end{pmatrix};\end{bmatrix}\right)$. Verify that $\mathsf{P}_1 + \mathsf{P}_2 + \mathsf{P}_3 = \mathsf{I}$. The basis $\boldsymbol{a}_1$, $\boldsymbol{a}_2$, $\boldsymbol{a}_3$ is orthogonal!

(Strang, 2003, exercise 5–7 from section 4.2.)

(L-13) Question 7. Project $\boldsymbol{b}$ onto the column space of A by solving $\mathsf{A}^\mathsf{T}\mathsf{A}\widehat{\boldsymbol{x}} = \mathsf{A}^\mathsf{T}\boldsymbol{b}$ and then computing $\widehat{\boldsymbol{b}} = \mathsf{A}\widehat{\boldsymbol{x}}$. Find $\boldsymbol{e} = \boldsymbol{b} - \widehat{\boldsymbol{b}}$.

(a) $\mathsf{A}_1 = \begin{bmatrix}1 & 1\\0 & 1\\0 & 0\end{bmatrix}$ and $\boldsymbol{b}_1 = \begin{pmatrix}2\\3\\4\end{pmatrix}$

(b) $\mathsf{A}_2 = \begin{bmatrix}1 & 1\\1 & 1\\0 & 1\end{bmatrix}$ and $\boldsymbol{b}_2 = \begin{pmatrix}4\\4\\6\end{pmatrix}$

(c) Compute the projection matrices P_1 and P_2 onto the column spaces. Verify that $\mathsf{P}_1\boldsymbol{b}_1$ gives the first projection $\widehat{\boldsymbol{b}_1}$. Also verify $(\mathsf{P}_2)^2 = \mathsf{P}_2$.

(Strang, 2003, exercise 11–12 from section 4.2.)

End of Questions of the Lecture 13

References

Hefferon, J. (2008). *Linear Algebra*. Jim Hefferon, Colchester, Vermont USA. This text is Free. URL ftp://joshua.smcvt.edu/pub/hefferon/book/book.pdf

Lang, S. (1986). *Introduction to Linear Algebra*. Springer-Verlag, second ed.

Strang, G. (2003). *Introduction to Linear Algebra*. Wellesley-Cambridge Press, Wellesley, Massachusetts. USA, third ed. ISBN 0-9614088-9-8.

Strang, G. (2006). *Linear algebra and its applications*. Thomson Learning, Inc., fourth ed. ISBN 0-03-010567-6.

Solutions

(L-11) Question 1. Since $\mathcal{C}\left(\mathbf{A}^\intercal\right) \perp \mathcal{N}\left(\mathbf{A}\right)$, we only need to find the orthogonal complement of the span of $\begin{bmatrix}1 & 3 & -1\end{bmatrix}$.

$$\left[\begin{array}{c}\mathbf{A}\\\hline\mathbf{I}\end{array}\right] = \left[\begin{array}{ccc}1&3&-1\\\hline 1&0&0\\0&1&0\\0&0&1\end{array}\right] \xrightarrow[\left[(1)\mathbf{1}+\mathbf{3}\right]]{\overset{\boldsymbol{\tau}}{\left[(-3)\mathbf{1}+\mathbf{2}\right]}} \left[\begin{array}{ccc}1&0&0\\\hline 1&-3&1\\0&1&0\\0&0&1\end{array}\right] = \left[\begin{array}{c}\mathbf{L}\\\hline\mathbf{E}\end{array}\right]$$

Therefore, the set of vectors in $\mathbb{R}^3$ orthogonal to $\begin{pmatrix}1, & 3, & -1\end{pmatrix}$ is

$$\left\{ \boldsymbol{v} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^2 \text{ tal que } \boldsymbol{v} = \begin{bmatrix}-3 & 1\\ 1 & 0\\ 0 & 1\end{bmatrix} \boldsymbol{p} \right\}.$$

With NAcAL there are several ways to obtain such a subspace. There are two ways to invoke `Subspace`: if the argument is a system (`Sistema`) of `Vectors` of $\mathbb{R}^n$, it returns the subspace spanned by that system.

```
a = Vector([-3,1,0])
b = Vector([1,0,1])
SubEspacio(Sistema([a,b]))
```

If the argument is a `Matrix`, it returns its null space.

```
v = Vector([1,3,-1])
A = ~Matrix([v])      # trasponemos para obtener la matriz fila
SubEspacio(A)
```

But since we are asked for the orthogonal complement of the subspace generated by the vector, we can simply write (since in this context means the orthogonal complement):

```
~SubEspacio(Sistema([v]))
```

The representation by means of parametric or Cartesian equations is not unique; in fact, we obtain different parametric equations for the systems $\left[\boldsymbol{a};\boldsymbol{b};\right]$ (seen above) and $\left[\boldsymbol{b};\boldsymbol{a};\right]$.

```
SubEspacio(Sistema([b,a]))
```

It is therefore useful to be able to check whether two subspaces are equal

```
~SubEspacio(Sistema([v])) == SubEspacio(Sistema([b,a]))
```

□

(L-11) Question 2. $\boldsymbol{a}\cdot\boldsymbol{a} = \sum_{i=1}^{n} a_i^2 = 0 \iff \boldsymbol{a} = \mathbf{0}$. Therefore, the answer is yes: the zero vector $\mathbf{0}$.

□

(L-11) Question 3(a) $\sqrt{3^2+1^2} = \sqrt{10}$

□

(L-11) Question 3(b) $\sqrt{5}$

□

(L-11) Question 3(c) $\sqrt{18}$

□

(L-11) Question 3(d) 0

□

(L-11) Question 3(e) $\sqrt{3}$

□

(L-11) Question 4. $\|\boldsymbol{v}\|^2 = \boldsymbol{v}\cdot\boldsymbol{v} = 4+1+0+16+4 = 25$ so we take $\boldsymbol{u} = \frac{1}{\|\boldsymbol{v}\|}\cdot\boldsymbol{v} = \left(\frac{2}{5}, \frac{-1}{5}, 0, \frac{4}{5}, \frac{-2}{5}\right)$.

□

(L-11) Question 5. Its dot product must be zero; therefore $(k)(4)+(1)(3)=0$ therefore $k=-\frac{3}{4}$.

□

(L-11) Question 6(a) $\begin{bmatrix}1&2&a\\2&-3&b\\-3&5&c\end{bmatrix}\begin{pmatrix}1\\1\\1\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix} \Rightarrow \begin{pmatrix}1\\2\\-3\end{pmatrix}+\begin{pmatrix}2\\-3\\5\end{pmatrix}+\begin{pmatrix}a\\b\\c\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix} \Rightarrow \begin{pmatrix}a\\b\\c\end{pmatrix}=\begin{pmatrix}-3\\1\\-2\end{pmatrix}$;
Thus, $\mathbf{A}=\begin{bmatrix}1&2&-3\\2&-3&1\\-3&5&-2\end{bmatrix}$.

□

(L-11) Question 6(b) Impossible, $\begin{pmatrix}2\\-3\\5\end{pmatrix}$ is not orthogonal to $\begin{pmatrix}1\\1\\1\end{pmatrix}$

□

(L-11) Question 6(c) $\begin{pmatrix}1\\1\\1\end{pmatrix}$ is in $\mathcal{C}\left(\mathbf{A}\right)$, and $\begin{pmatrix}1\\0\\0\end{pmatrix}$ is in $\mathcal{N}\left(\mathbf{A}^{\mathsf{T}}\right)$. It is impossible: these vectors are not orthogonal.

□

(L-11) Question 6(d) This asks for $\mathbf{A}\cdot\mathbf{A}=\mathbf{0}$. For example $\mathbf{A}=\begin{bmatrix}1&-1\\1&-1\end{bmatrix}$, or $\mathbf{A}=\begin{bmatrix}0&0\\1&0\end{bmatrix}$ for example.

□

(L-11) Question 6(e) $\begin{pmatrix}1, & 1, & 1,\end{pmatrix}$ must be in the null space, $\mathbf{A}\begin{pmatrix}1\\1\\1\end{pmatrix}=\mathbf{0}$; and row space, $\begin{pmatrix}1, & 1, & 1,\end{pmatrix}\mathbf{A}=\begin{pmatrix}1, & 1, & 1,\end{pmatrix}$, ... such a matrix does not exist.

□

(L-11) Question 7. If $\mathbf{AB}=\mathbf{0}$, the columns of $\mathbf{B}$ are in the *null space* of $\mathbf{A}$. The rows of $\mathbf{A}$ are in the *left null space* of $\mathbf{B}$.

If rank = 2, all four subspaces would have dimension 2 which is impossible for a 3 by 3 matrix.

□

(L-11) Question 8. No. For example.

$$\boldsymbol{u}=\begin{pmatrix}1\\0\end{pmatrix} \quad \boldsymbol{v}=\begin{pmatrix}1\\0\end{pmatrix} \quad \boldsymbol{w}=\begin{pmatrix}1\\1\end{pmatrix},$$

$\boldsymbol{u}\cdot\boldsymbol{v}=1=\boldsymbol{u}\cdot\boldsymbol{w}$, but $\boldsymbol{v}\neq\boldsymbol{w}$.

□

(L-11) Question 9(a) On the one hand $\mathbf{A}\boldsymbol{x}=\boldsymbol{b}\Rightarrow\boldsymbol{b}\in\mathcal{C}\left(\mathbf{A}\right)$ and on the other hand $\mathbf{A}^{\mathsf{T}}\boldsymbol{y}=\mathbf{0}\Rightarrow\boldsymbol{y}\perp\in\mathcal{C}\left(\mathbf{A}\right)$.

If $\mathbf{A}\boldsymbol{x}=\boldsymbol{b}$ has a solution and $\mathbf{A}^{\mathsf{T}}\boldsymbol{y}=\mathbf{0}$, then $\boldsymbol{y}$ is perpendicular to $\boldsymbol{b}$.

$$\boldsymbol{y}\cdot\boldsymbol{b}=\boldsymbol{y}\mathbf{A}\boldsymbol{b}=\mathbf{0}\cdot\boldsymbol{b}=0.$$

□

(L-11) Question 9(b) If $\mathbf{A}^{\mathsf{T}}\boldsymbol{y}=\boldsymbol{c}$ then $\boldsymbol{y}\mathbf{A}=\boldsymbol{c}$, also $\mathbf{A}\boldsymbol{x}=\mathbf{0}$; then $\boldsymbol{x}$ is perpendicular to $\boldsymbol{c}$.

$\boldsymbol{c}$ is in the row space, and therefore it is orthogonal to $\boldsymbol{x}$, which is a vector in the null space. In other words:

$$\boldsymbol{c}\cdot\boldsymbol{x}=\boldsymbol{y}\mathbf{A}\boldsymbol{x}=\boldsymbol{y}\cdot\mathbf{0}=0.$$

□

(L-11) Question 10. If $\boldsymbol{u}$ and $\boldsymbol{v}$ are perpendicular then

$$\|(\boldsymbol{u}+\boldsymbol{v})\|^2 = (\boldsymbol{u}+\boldsymbol{v})\cdot(\boldsymbol{u}+\boldsymbol{v}) = \boldsymbol{u}\cdot\boldsymbol{u}+2\big(\boldsymbol{u}\cdot\boldsymbol{v}\big)+\boldsymbol{v}\cdot\boldsymbol{u} = \boldsymbol{u}\cdot\boldsymbol{u}+\boldsymbol{v}\cdot\boldsymbol{v} = \|\boldsymbol{u}\|^2+\|\boldsymbol{v}\|^2$$

(the third equality holds because $\boldsymbol{u}\cdot\boldsymbol{v}=0$).

□

(L-11) Question 11(a) Since $\mathcal{N}\left(\mathbf{A}\right) \perp \mathcal{C}\left(\mathbf{A}^{\intercal}\right)$, any 1 by 3 matrix whose row is a non-zero multiple of $\begin{pmatrix}1, & 2, & 4,\end{pmatrix}$ is a correct answer. For example, $\left[\begin{array}{ccc} 1 & 2 & 4 \end{array}\right]$.

To obtain a 3 by 3 matrix, simply add two rows that are multiples of the first row to the previous matrix. For example, $\left[\begin{array}{ccc} 1 & 2 & 4 \\ 1 & 2 & 4 \\ 1 & 2 & 4 \end{array}\right]$.

□

(L-11) Question 11(b) We can take as a row of $\mathbf{A}$, a linear combination of a basis of the left null space of $\left[\begin{array}{c} 1 \\ 2 \\ 4 \end{array}\right]$.

Thus,

$$\left[\begin{array}{ccc} 1 & 2 & 4 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[(-4)\mathbf{1}+\mathbf{3}]}]{\boldsymbol{\tau}\substack{[(-2)\mathbf{1}+\mathbf{2}]}} \left[\begin{array}{ccc} 1 & 0 & 0 \\ \hline 1 & -2 & -4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \quad \Rightarrow \quad \left[\begin{array}{ccc} -2 & 1 & 0 \\ -4 & 0 & 1 \end{array}\right]$$

and then

$$\underset{1\times 3}{\mathbf{A}} = \left[\begin{array}{cc} 1 & 1 \end{array}\right] \left[\begin{array}{ccc} -2 & 1 & 0 \\ -4 & 0 & 1 \end{array}\right] = \left[\begin{array}{ccc} -6 & 1 & 1 \end{array}\right]$$

satisfies the requirement. A 3 by 3 matrix with that same null space is, for instance

$$\mathbf{C} = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{array}\right] \left[\begin{array}{ccc} -2 & 1 & 0 \\ -4 & 0 & 1 \end{array}\right] = \left[\begin{array}{ccc} -2 & 1 & 0 \\ -4 & 0 & 1 \\ -6 & 1 & 1 \end{array}\right]$$

has the same null space...

□

(L-11) Question 12(a) Any column of $\mathbf{A}$ is orthogonal to the two special solutions given in the problem. That is,

$$\left[\begin{array}{cccc} 3 & 1 & 0 & 0 \\ 6 & 0 & 2 & 1 \\ \hline 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[(-6)\mathbf{4}+\mathbf{1}]\\[(-2)\mathbf{4}+\mathbf{3}]}]{\boldsymbol{\tau}\substack{[(-3)\mathbf{2}+\mathbf{1}]}} \left[\begin{array}{cccc} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ \hline 1 & 0 & 0 & 0 \\ -3 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -6 & 0 & -2 & 1 \end{array}\right] \quad \text{therefore} \quad \mathbf{R} = \left[\begin{array}{cc} 1 & 0 \\ -3 & 0 \\ 0 & 1 \\ -6 & -2 \end{array}\right].$$

□

(L-11) Question 12(b) $\mathbf{R}$ has two pivots, and therefore $\mathbf{A}$ has two pivots and $r(\mathbf{A}) = 2$. Two independent rows in $\mathbb{R}^2$ span $\mathbb{R}^2$, so $\mathcal{C}\left(\mathbf{A}^{\intercal}\right) = \mathbb{R}^2$.

□

(L-11) Question 12(c) Since rows 1 and 3 are pivot rows, then $\boldsymbol{x}_p = \begin{pmatrix}3, & 0, & 6, & 0,\end{pmatrix}$ is a particular solution, so the complete solution is

$$\left\{\boldsymbol{v} \in \mathbb{R}^4 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^2,\ \boldsymbol{v} = \begin{pmatrix}3, & 0, & 6, & 0,\end{pmatrix} + \boldsymbol{p}\left[\begin{array}{cccc} 3 & 1 & 0 & 0 \\ 6 & 0 & 2 & 1 \end{array}\right]\right\}$$

since

$$\begin{pmatrix}3, & 0, & 6, & 0,\end{pmatrix}\left[\begin{array}{cc} 1 & 0 \\ -3 & 0 \\ 0 & 1 \\ -6 & -2 \end{array}\right] = \begin{pmatrix}3, & 6,\end{pmatrix}$$

and

$$\boldsymbol{p}\left[\begin{array}{cccc} 3 & 1 & 0 & 0 \\ 6 & 0 & 2 & 1 \end{array}\right]\left[\begin{array}{cc} 1 & 0 \\ -3 & 0 \\ 0 & 1 \\ -6 & -2 \end{array}\right] = \boldsymbol{p}\left[\begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array}\right] = \begin{pmatrix}0, & 0,\end{pmatrix}.$$

□

(L-11) Question 12(d) It is easy to see that

$$-2\begin{pmatrix}-3\\0\end{pmatrix}+2\begin{pmatrix}0\\1\end{pmatrix}+\begin{pmatrix}-6\\-2\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}$$

If you don't see that, we can always use Gaussian elimination

$$\left[\begin{array}{ccc}-3&0&-6\\0&1&-2\\\hline 1&0&0\\0&1&0\\0&0&1\end{array}\right]\xrightarrow{[(-2)\boldsymbol{\tau}\mathbf{1}+\mathbf{3}]}\left[\begin{array}{ccc}-3&0&0\\0&1&-2\\\hline 1&0&-2\\0&1&0\\0&0&1\end{array}\right]\xrightarrow{[(2)\boldsymbol{\tau}\mathbf{2}+\mathbf{3}]}\left[\begin{array}{ccc}-3&0&0\\0&1&0\\\hline 1&0&-2\\0&1&2\\0&0&1\end{array}\right].$$

□

(L-11) Question 13(a)

$$\boldsymbol{b}\;=\;\mathbf{A}\boldsymbol{x}\;=\;\mathbf{A}\big(\boldsymbol{x}_r+\boldsymbol{x}_n\big)\;=\;\mathbf{A}(\boldsymbol{x}_r)+\mathbf{A}(\boldsymbol{x}_n)\;=\;\mathbf{A}(\boldsymbol{x}_r)+\mathbf{0}.$$

□

(L-11) Question 13(b) If $\mathbf{A}(\boldsymbol{x}_r)=\boldsymbol{b}$ and $\mathbf{A}(\boldsymbol{v}_r)=\boldsymbol{b}$, then $\mathbf{A}\big(\boldsymbol{x}_r-\boldsymbol{v}_r\big)=\boldsymbol{b}-\boldsymbol{b}=\mathbf{0}$.

Therefore, $(\boldsymbol{x}_r-\boldsymbol{v}_r)$ belongs simultaneously to $\mathcal{N}\left(\mathbf{A}\right)$ and $\mathcal{C}\left(\mathbf{A}^\intercal\right)$ (since both $\boldsymbol{x}_r$ and $\boldsymbol{v}_r$ are linear combinations of the rows of $\mathbf{A}$).

Since $\mathcal{N}\left(\mathbf{A}\right)$ is perpendicular to $\mathcal{C}\left(\mathbf{A}^\intercal\right)$, the zero vector is the only vector that is simultaneously a solution of $\mathbf{A}\boldsymbol{x}=\mathbf{0}$ and a linear combination of the rows of $\mathbf{A}$. Therefore, $(\boldsymbol{x}_r-\boldsymbol{v}_r)=\mathbf{0}$. In other words, $\boldsymbol{x}_r$ and $\boldsymbol{v}_r$ are equal.

□

(L-11) Question 13(c) Let's solve $\begin{bmatrix}1&2&3\\1&1&-1\end{bmatrix}\boldsymbol{x}_r=\begin{pmatrix}14\\9\end{pmatrix}$. Then $\boldsymbol{x}_r=\begin{bmatrix}1&1\\2&1\\3&-1\end{bmatrix}\begin{pmatrix}c\\d\end{pmatrix}$, then

$$\begin{bmatrix}1&2&3\\1&1&-1\end{bmatrix}\boldsymbol{x}_r\;=\;\begin{bmatrix}1&2&3\\1&1&-1\end{bmatrix}\begin{bmatrix}1&1\\2&1\\3&-1\end{bmatrix}\begin{pmatrix}c\\d\end{pmatrix}\;=\;\begin{pmatrix}14\\9\end{pmatrix}$$

Hence $\begin{bmatrix}14&0\\0&3\end{bmatrix}\begin{pmatrix}c\\d\end{pmatrix}=\begin{pmatrix}14\\9\end{pmatrix}$, which implies $c=1$ and $d=3$. Thus $\boldsymbol{x}_r=1\begin{pmatrix}1\\2\\3\end{pmatrix}+3\begin{pmatrix}1\\1\\-1\end{pmatrix}=\begin{pmatrix}4\\5\\0\end{pmatrix}$.

□

(L-12) Question 1(a) Let us first find a vector parallel to the line. We define

$$\boldsymbol{v}=\boldsymbol{x}_P-\boldsymbol{x}_Q=\begin{pmatrix}1\\2\end{pmatrix}+\begin{pmatrix}3\\1\end{pmatrix}=\begin{pmatrix}-2\\1\end{pmatrix}.$$

Therefore, a parametric representation of the line is

$$\left\{\boldsymbol{v}\in\mathbb{R}^2\;\middle|\;\exists\boldsymbol{p}\in\mathbb{R}^1,\;\boldsymbol{v}=\begin{pmatrix}1\\2\end{pmatrix}+\begin{bmatrix}-2\\1\end{bmatrix}\boldsymbol{p}\right\}.$$

With NAcAL, points, lines, planes, etc. (i.e., planar regions in R[n]) are created using `EAfin`. The required arguments for `EAfin` are a `Subspace` and a `Vector`. If instead of a `Subspace` a `System` of `Vectors` in $\mathbb{R}^n$ or a `Matrix` is provided, NAcAL will use those arguments to generate the necessary subspace (the subspace generated by the system in the first case or the null space of the matrix in the second).

Thus, in this case we obtain the equations of the required line with:

```
p = Vector([1,2])
q = Vector([3,1])
S = SubEspacio(Sistema([p-q]))
R = EAfin(S,p)
Math( R.EcParametricas() )  # Por ahora solo quiero visualizar las Ec. Paramétricas de R
```

□

(L-12) Question 1(b) We need to multiply $\boldsymbol{x} = \boldsymbol{x}_P + a\boldsymbol{v}$ by a vector perpendicular to $\boldsymbol{v}$. We will accomplish this via Gaussian elimination:

$$\left[\begin{array}{cc} -2 & 1 \\ \hline x & y \\ \hline 1 & 2 \end{array}\right] \xrightarrow[[(1)\mathbf{1}+\mathbf{2}]]{\overset{\boldsymbol{\tau}}{[(2)\mathbf{2}]}} \left[\begin{array}{cc} -2 & 0 \\ \hline x & x+2y \\ \hline 1 & 5 \end{array}\right] \Rightarrow \text{ the solution set of } \{x+2y=5\,;$$

and therefore the line is:

$$\left\{\boldsymbol{v} \in \mathbb{R}^2 \;\middle|\; \begin{bmatrix} 1 & 2 \end{bmatrix} \boldsymbol{v} = (5,)\right\}.$$

Let's reproduce the pencil and paper calculation with NAcAL.

```
x,y = sympy.symbols('x y')
N = ~Matrix([p-q])
M = N.apila(~Matrix([Vector([x,y])]),1).apila(~Matrix([p]),1)
Math( rprElim(M, Elim(N).pasos) )
```

Therefore, the straight line is the set of vectors that solve the following system of linear equations:

```
A = Matrix([[1,2]])
b = Vector([5])
SEL(A,b).eafin
```

(note that NAcAL stores as an attribute (of type `EAfin`) the set of solutions to a system of equations)

□

(L-12) Question 2(a) Let us first find a vector in the same direction as the line. We define

$$\boldsymbol{v} = \boldsymbol{x}_P - \boldsymbol{x}_Q = \begin{pmatrix} 1 \\ -3 \\ 1 \end{pmatrix} - \begin{pmatrix} -2 \\ 4 \\ 5 \end{pmatrix} = (3, \quad -7, \quad -4,).$$

Therefore, a parametric representation of the line is

$$\left\{\boldsymbol{v} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^1,\ \boldsymbol{v} = (1, \quad -3, \quad 1,) + \begin{bmatrix} 3 \\ -7 \\ -4 \end{bmatrix} \boldsymbol{p}\right\}.$$

□

(L-12) Question 2(b)

$$\left[\begin{array}{ccc} 3 & -7 & -4 \\ \hline x & y & z \\ \hline 1 & -3 & 1 \end{array}\right] \xrightarrow[[(7)\mathbf{1}+\mathbf{2}]]{\overset{\boldsymbol{\tau}}{[(3)\mathbf{2}]}} \xrightarrow[[(4)\mathbf{1}+\mathbf{3}]]{[(3)\mathbf{3}]} \left[\begin{array}{ccc} 3 & 0 & 0 \\ \hline x & 7x+3y & 4x+3z \\ \hline 1 & -2 & 7 \end{array}\right] \Rightarrow \begin{cases} 7x+3y & = -2 \\ 4x & +3z = 7 \end{cases};$$

Therefore, the Cartesian equations of the line are:

$$\left\{\boldsymbol{v} \in \mathbb{R}^3 \;\middle|\; \begin{bmatrix} 7 & 3 & 0 \\ 4 & 0 & 3 \end{bmatrix} \boldsymbol{v} = \begin{pmatrix} -2 \\ 7 \end{pmatrix}\right\}.$$

This system has two equations. If we take them separately, they correspond to two planes in $\mathbb{R}^3$.

```
p1=SEL(Matrix([[7,3,0]]),Vector([-2])).eafin
p1
```

and

```
p1=SEL(Matrix([[7,3,0]]),Vector([-2])).eafin
p1
```

(we know that they are two planes, since the parametric equations have two parameters, and the coefficient matrices of the Cartesian equations have two free columns) The line in this exercise corresponds to the intersection of both planes, that is, to the points that belong to both planes:

```
p1 & p2
```

□

(L-12) Question 3(a) Since it is parallel to the line $2x - 3y = 5$, we need to find a vector $\boldsymbol{v}$ in the nullspace of the coeficient matrix of the system, for example:

$$\left[\begin{array}{cc} 2 & -3 \\ \hline 1 & 0 \\ 0 & 1 \end{array}\right] \xrightarrow[\substack{[(6)\mathbf{1}] \\ [(2)\mathbf{2}]}]{\boldsymbol{\tau}} \left[\begin{array}{cc} 6 & -6 \\ \hline 3 & 0 \\ 0 & 2 \end{array}\right] \xrightarrow{\boldsymbol{\tau}_{[(1)\mathbf{2}+\mathbf{1}]}} \left[\begin{array}{cc} 6 & 0 \\ \hline 3 & 3 \\ 0 & 2 \end{array}\right]$$

therefore $\left\{ \boldsymbol{x} \in \mathbb{R}^2 \;\middle|\; \exists a \in \mathbb{R} \text{ tal que } \boldsymbol{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + a \begin{pmatrix} 3 \\ 2 \end{pmatrix} \right\}$

□

(L-12) Question 3(b) We only need to substitute (x, y) by $(1, 1)$ to obtain the right hand side "vector" $\boldsymbol{b}$.

$$2x - 3y = 2 \cdot 1 - 3 \cdot 1 = -1 \quad \Rightarrow \quad 2x - 3y = -1.$$

Hence

$$\left\{ \begin{pmatrix} x \\ y \end{pmatrix} \;\middle|\; \begin{bmatrix} 2 & -3 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = (-1) \right\}$$

□

(L-12) Question 4(a) $\left\{ \boldsymbol{v} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^2,\ \boldsymbol{v} = \begin{pmatrix} 0, & 1, & 1, \end{pmatrix} + \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 0 \end{bmatrix} \boldsymbol{p} \right\}.$

□

(L-12) Question 4(b)

$$\left[\begin{array}{ccc} 0 & 1 & 2 \\ 1 & 1 & 0 \\ \hline x & y & z \\ \hline 0 & 1 & 1 \end{array}\right] \xrightarrow{\boldsymbol{\tau}_{[(-1)\mathbf{1}+\mathbf{2}]}} \left[\begin{array}{ccc} 0 & 1 & 2 \\ 1 & 0 & 0 \\ \hline x & -x+y & z \\ \hline 0 & 1 & 1 \end{array}\right] \xrightarrow{\boldsymbol{\tau}_{[(-2)\mathbf{2}+\mathbf{3}]}} \left[\begin{array}{ccc} 0 & 1 & 0 \\ 1 & 0 & 0 \\ \hline x & -x+y & 2x-2y+z \\ \hline 0 & 1 & -1 \end{array}\right]$$

Therefore: $\left\{ \boldsymbol{v} \in \mathbb{R}^3 \;\middle|\; \begin{bmatrix} 2 & -2 & 1 \end{bmatrix} \boldsymbol{v} = (-1,) \right\}.$

```
p = Vector([0,1,1])
v = Vector([0,1,2])
w = Vector([1,1,0])
S = SubEspacio(Sistema([v,w]))
EAfin(S,p)
```

□

(L-12) Question 5(a) Since we are asked for a plane in $\mathbb{R}^3$, we need to find two vectors orthogonal to $\begin{pmatrix} 3, & 1, & 1, \end{pmatrix}$. For example, $\begin{pmatrix} -1, & 3, & 0, \end{pmatrix}$ and $\begin{pmatrix} 0, & -1, & 1, \end{pmatrix}$. Therefore,

$$\left\{ \boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^2 \text{ tal que } \boldsymbol{x} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \begin{bmatrix} 1 & 0 \\ -3 & -1 \\ 0 & 1 \end{bmatrix} \boldsymbol{p} \right\}.$$

□

(L-12) Question 5(b) In this case, we already know from the problem statement a vector orthogonal to the parametric part; thus:

$$\begin{bmatrix}3 & 1 & 1\end{bmatrix}\boldsymbol{x} = \begin{bmatrix}3 & 1 & 1\end{bmatrix}\boldsymbol{s}; \quad \Rightarrow \quad \begin{bmatrix}3 & 1 & 1\end{bmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix} = \begin{bmatrix}3 & 1 & 1\end{bmatrix}\begin{pmatrix}2\\1\\3\end{pmatrix} = (10,);$$

$$\Longrightarrow \qquad \left\{\boldsymbol{x}\in\mathbb{R}^2 \;\middle|\; \begin{bmatrix}3 & 1 & 1\end{bmatrix}\boldsymbol{x} = (10,)\right\}.$$

```
p = Vector([2,1,3])
v = Vector([3,1,1])
S =  SubEspacio(~Matrix([v])) # esta es una alternativa
#S = ~SubEspacio(Sistema([v])) # esta es otra alternativa
EAfin(S,p)
```

□

(L-12) Question 6(a) La solución completa es:

$$\boldsymbol{b} = \left\{ \boldsymbol{v}\in\mathbb{R}^5 \;\middle|\; \exists \boldsymbol{p}\in\mathbb{R}^1,\ \boldsymbol{v} = \begin{pmatrix}-1\\0\\1\\-2\\4\end{pmatrix} + \begin{bmatrix}-2\\1\\0\\0\\0\end{bmatrix}\boldsymbol{p} \right\}$$

```
A = Matrix([ [1,2,0,1,1], [0,0,2,3,1], [0,0,1,4,2], [0,0,0,1,1] ])
b = Vector( [1,0,1,2] )
SEL (A, b, 1)
```

□

(L-12) Question 6(b) Since the coefficient matrix has five columns, the system has five unknowns. Thus, the vectors that belong to the solution set have five components (one number per column). Therefore, the solution set is a subset of $\mathbb{R}^5$; and in this case, that set is a line, since the dimension of $\mathcal{N}\left(\mathbf{A}\right)$ is one. Hence, a direction vector is any multiple (except the zero vector $\mathbf{0}$) of the special solution we have found: $\boldsymbol{n} = \begin{pmatrix}-2, & 1, & 0, & 0, & 0,\end{pmatrix}$. And one of the points through which the line passes is the particular solution we obtained by solving the system: $\boldsymbol{s} = \begin{pmatrix}-1, & 0, & 1, & -2, & 4,\end{pmatrix}$.

□

(L-12) Question 6(c)

$$\left[\begin{array}{c}\left[\boldsymbol{n}\right]^{\intercal}\\\hline \mathbf{I}\end{array}\right] = \left[\begin{array}{ccccc|c} -2 & 1 & 0 & 0 & 0 & \\ \hline 1 & 0 & 0 & 0 & 0 & \\ 0 & 1 & 0 & 0 & 0 & \\ 0 & 0 & 1 & 0 & 0 & \\ 0 & 0 & 0 & 1 & 0 & \\ 0 & 0 & 0 & 0 & 1 & \end{array}\right] \xrightarrow[[(1)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}_{[(2)\mathbf{2}]}} \left[\begin{array}{ccccc|c} -2 & 0 & 0 & 0 & 0 & \\ \hline 1 & 1 & 0 & 0 & 0 & \\ 0 & 2 & 0 & 0 & 0 & \\ 0 & 0 & 1 & 0 & 0 & \\ 0 & 0 & 0 & 1 & 0 & \\ 0 & 0 & 0 & 0 & 1 & \end{array}\right] = \left[\begin{array}{c}\mathbf{L}\\\hline \mathbf{E}\end{array}\right]$$

The last four columns of the matrix $\mathbf{E}$ are vectors perpendicular to $\boldsymbol{n}$; it is evident that there are four of them and that they are linearly independent, so they form a basis of the subspace perpendicular to $\boldsymbol{n}$.

□

(L-13) Question 1(a)

$$\widehat{\boldsymbol{b}} = \left[\boldsymbol{a}\right]\left(\left[\boldsymbol{a}\right]^{\intercal}\left[\boldsymbol{a}\right]\right)^{-1}\left[\boldsymbol{a}\right]^{\intercal}\boldsymbol{b} = \begin{bmatrix}3\\2\end{bmatrix}\left(\begin{bmatrix}3 & 2\end{bmatrix}\begin{bmatrix}3\\2\end{bmatrix}\right)^{-1}\begin{bmatrix}3 & 2\end{bmatrix}\begin{pmatrix}2\\1\end{pmatrix} = \frac{1}{13}\begin{bmatrix}3\\2\end{bmatrix}\begin{bmatrix}3 & 2\end{bmatrix}\begin{pmatrix}2\\1\end{pmatrix} = \frac{1}{13}\begin{pmatrix}24\\16\end{pmatrix}$$

$$\boldsymbol{e} = \boldsymbol{b} - \widehat{\boldsymbol{b}} = \begin{pmatrix}2\\1\end{pmatrix} - \frac{1}{13}\begin{pmatrix}24\\16\end{pmatrix} = \frac{1}{13}\begin{pmatrix}2\\-3\end{pmatrix}$$

$$\boldsymbol{a}\cdot\boldsymbol{e} = \begin{pmatrix}3, & 2,\end{pmatrix}\begin{pmatrix}2\\-3\end{pmatrix}\frac{1}{13} = 0\frac{1}{13} = 0.$$

$$\mathbf{P} = \frac{1}{13}\begin{bmatrix}9 & 6\\6 & 4\end{bmatrix}; \quad \mathbf{P}^2 = \frac{1}{13}\cdot\frac{1}{13}\begin{bmatrix}117 & 78\\78 & 52\end{bmatrix} = \frac{1}{13}\begin{bmatrix}9 & 6\\6 & 4\end{bmatrix} = \mathbf{P};$$

$$\mathbf{P}\boldsymbol{b} = \frac{1}{13}\begin{bmatrix} 9 & 6 \\ 6 & 4 \end{bmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \frac{1}{13}\begin{pmatrix} 24 \\ 16 \end{pmatrix}.$$

```
a      = Vector([3,2]);  b = Vector([2,1]);   A = Matrix([a])
P      = A*InvMat((~A)*A)*(~A)       # Matriz proyección
bhat   = P*b                         # Alternativa 1
e      = b-bhat
Sistema([bhat,e,P])
```

□

(L-13) Question 1(b)

$$\widehat{\boldsymbol{b}} = [\boldsymbol{a}]\left([\boldsymbol{a}]^{\intercal}[\boldsymbol{a}]\right)^{-1}[\boldsymbol{a}]^{\intercal}\boldsymbol{b} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}\left(\begin{bmatrix} 3 & 0 \end{bmatrix}\begin{bmatrix} 3 \\ 0 \end{bmatrix}\right)^{-1}\begin{bmatrix} 3 & 0 \end{bmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \frac{1}{9}\begin{bmatrix} 3 \\ 0 \end{bmatrix}\begin{bmatrix} 3 & 0 \end{bmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \frac{1}{9}\begin{pmatrix} 18 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$

$$\boldsymbol{e} = \boldsymbol{b} - \widehat{\boldsymbol{b}} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} - \frac{1}{9}\begin{pmatrix} 18 \\ 0 \end{pmatrix} = \frac{1}{9}\begin{pmatrix} 0 \\ 9 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\boldsymbol{a}\cdot\boldsymbol{e} = \begin{pmatrix} 3, & 0, \end{pmatrix}\begin{pmatrix} 0 \\ 9 \end{pmatrix}\frac{1}{9} = 0\frac{1}{9} = 0.$$

$$\mathbf{P} = \frac{1}{9}\begin{bmatrix} 9 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix};\quad \mathbf{P}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \mathbf{P};$$

$$\mathbf{P}\boldsymbol{b} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$

□

(L-13) Question 1(c)

$$\widehat{\boldsymbol{b}} = [\boldsymbol{a}]\left([\boldsymbol{a}]^{\intercal}[\boldsymbol{a}]\right)^{-1}[\boldsymbol{a}]^{\intercal}\boldsymbol{b} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}\left(\begin{bmatrix} 1 & 2 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}\right)^{-1}\begin{bmatrix} 1 & 2 & -1 \end{bmatrix}\begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} =$$

$$\frac{1}{6}\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}\begin{bmatrix} 1 & 2 & -1 \end{bmatrix}\begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}$$

$$\boldsymbol{e} = \boldsymbol{b} - \widehat{\boldsymbol{b}} = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} - \frac{1}{6}\begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 7 \\ 8 \\ 23 \end{pmatrix}$$

$$\boldsymbol{a}\cdot\boldsymbol{e} = \begin{pmatrix} 1, & 2, & -1, \end{pmatrix}\begin{pmatrix} 7 \\ 8 \\ 23 \end{pmatrix}\frac{1}{6} = 0\frac{1}{6} = 0.$$

$$\mathbf{P} = \frac{1}{6}\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{bmatrix};\quad \mathbf{P}^2 = \frac{1}{6}\cdot\frac{1}{6}\begin{bmatrix} 6 & 12 & -6 \\ 12 & 24 & -12 \\ -6 & -12 & 6 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{bmatrix}.$$

$$\mathbf{P}\boldsymbol{b} = \frac{1}{6}\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{bmatrix}\begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}.$$

□

(L-13) Question 1(d)

$$\widehat{\boldsymbol{b}} = [\boldsymbol{a}]\left([\boldsymbol{a}]^{\intercal}[\boldsymbol{a}]\right)^{-1}[\boldsymbol{a}]^{\intercal}\boldsymbol{b} = \begin{bmatrix} 3 \\ 3 \\ 12 \end{bmatrix}\left(\begin{bmatrix} 3 & 3 & 12 \end{bmatrix}\begin{bmatrix} 3 \\ 3 \\ 12 \end{bmatrix}\right)^{-1}\begin{bmatrix} 3 & 3 & 12 \end{bmatrix}\begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} =$$

$$\frac{1}{162}\begin{bmatrix} 3 \\ 3 \\ 12 \end{bmatrix}\begin{bmatrix} 3 & 3 & 12 \end{bmatrix}\begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix}.$$

$$\boldsymbol{e} = \boldsymbol{b} - \widehat{\boldsymbol{b}} = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

$$\boldsymbol{a} \cdot \boldsymbol{e} = \boldsymbol{a} \cdot \boldsymbol{0} = 0.$$

$$\mathbf{P} = \frac{1}{18}\begin{bmatrix} 1 & 1 & 4 \\ 1 & 1 & 4 \\ 4 & 4 & 16 \end{bmatrix}; \quad \mathbf{P}^2 = \frac{1}{18}\frac{1}{18}\begin{bmatrix} 18 & 18 & 72 \\ 18 & 18 & 72 \\ 72 & 72 & 288 \end{bmatrix} = \mathbf{P};$$

$$\mathbf{P}\boldsymbol{b} = \frac{1}{18}\begin{bmatrix} 1 & 1 & 4 \\ 1 & 1 & 4 \\ 4 & 4 & 16 \end{bmatrix}\begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix}.$$

□

(L-13) Question 2(a)

$$\widehat{\boldsymbol{b}} = [\boldsymbol{a}]\left([\boldsymbol{a}]^{\intercal}[\boldsymbol{a}]\right)^{-1}[\boldsymbol{a}]^{\intercal}\boldsymbol{b} = \begin{bmatrix} -3 \\ 1 \\ -3 \end{bmatrix}\left(\begin{bmatrix} -3 & 1 & -3 \end{bmatrix}\begin{bmatrix} -3 \\ 1 \\ -3 \end{bmatrix}\right)^{-1}\begin{bmatrix} -3 & 1 & -3 \end{bmatrix}\begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} =$$

$$\frac{1}{19}\begin{bmatrix} -3 \\ 1 \\ -3 \end{bmatrix}\begin{bmatrix} -3 & 1 & -3 \end{bmatrix}\begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} = \frac{1}{19}\begin{bmatrix} 9 & -3 & 9 \\ -3 & 1 & -3 \\ 9 & -3 & 9 \end{bmatrix}\begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 3 \end{pmatrix}$$

```
b      = Vector([2,-1,4]); a = Vector([-3,1,-3]); A = Matrix([a])
P      = A*InvMat((~A)*A)*(~A)      # Matriz proyección
bhat1  = P*b                        # Alternativa 1
x      = SEL( (~A)*A, (~A)*b ).solP # Solución Ecuaciones Normales
bhat2  = A*x                        # Alternativa 2
Sistema([bhat1,bhat2])
```

□

(L-13) Question 2(b) The line is the set of solutions to $3x - y = 0$:

$$\left[\begin{array}{c} \mathbf{A} \\ \hline \mathbf{I} \end{array}\right] = \left[\begin{array}{cc} 3 & -1 \\ \hline 1 & 0 \\ 0 & 1 \end{array}\right] \xrightarrow[[(1)\mathbf{1}+\mathbf{2}]]{\overset{\boldsymbol{\tau}}{[(3)\mathbf{2}]}} \left[\begin{array}{cc} 3 & 0 \\ \hline 1 & 1 \\ 0 & 3 \end{array}\right] = \left[\begin{array}{c} \mathbf{L} \\ \hline \mathbf{E} \end{array}\right];$$

So we should project onto the line

$$\text{The line}: \left\{ \boldsymbol{v} \in \mathbb{R}^2 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^1,\ \boldsymbol{v} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}\boldsymbol{p} \right\}$$

$$\widehat{\boldsymbol{b}} = [\boldsymbol{a}]\left([\boldsymbol{a}]^{\intercal}[\boldsymbol{a}]\right)^{-1}[\boldsymbol{a}]^{\intercal}\boldsymbol{b} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}\left(\begin{bmatrix} 1 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right)^{-1}\begin{bmatrix} 1 & 3 \end{bmatrix}\begin{pmatrix} -1 \\ -1 \end{pmatrix} =$$

$$\frac{1}{10}\begin{bmatrix} 1 \\ 3 \end{bmatrix}\begin{bmatrix} 1 & 3 \end{bmatrix}\begin{pmatrix} -1 \\ -1 \end{pmatrix} = \frac{1}{10}\begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}\begin{pmatrix} -1 \\ -1 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} -2 \\ -6 \end{pmatrix};$$

```
b      = Vector([-1,-1])
B      = Matrix([[3,-1]])
```

```
a      = Homogenea(B).sgen|1
# a      = Homogenea(B).enulo.sgen|1 # alternativa equivalente
# a      = EAfin(B, V0(2)).S.sgen|1  # alternativa equivalente

A      = Matrix([a])
P      = A*InvMat((~A)*A)*(~A)       # Matriz proyección
bhat1  = P*b                         # Alternativa 1
x      = SEL( (~A)*A, (~A)*b ).solP # Solución Ecuaciones Normales
bhat2  = A*x                         # Alternativa 2
Sistema([bhat1, bhat2])
```

□

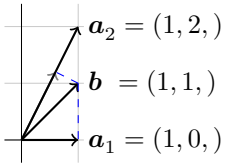
(L-13) Question 3.

$$\widehat{\boldsymbol{b}} = [\boldsymbol{a}]\left([\boldsymbol{a}]^{\intercal}[\boldsymbol{a}]\right)^{-1}[\boldsymbol{a}]^{\intercal}\boldsymbol{b} = \begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix}\left(\begin{bmatrix} -1 & 1 & -1 & 1 \end{bmatrix}\begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix}\right)^{-1}\begin{bmatrix} -1 & 1 & -1 & 1 \end{bmatrix}\begin{pmatrix} 1 \\ 2 \\ 1 \\ 3 \end{pmatrix} =$$

$$\frac{1}{4}\begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix}\begin{bmatrix} -1 & 1 & -1 & 1 \end{bmatrix}\begin{pmatrix} 1 \\ 2 \\ 1 \\ 3 \end{pmatrix} = \frac{1}{4}\begin{bmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{bmatrix}\begin{pmatrix} 1 \\ 2 \\ 1 \\ 3 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} -3 \\ 3 \\ -3 \\ 3 \end{pmatrix};$$

□

(L-13) Question 4(a) $\widehat{\boldsymbol{b}_1} = (1, \quad 0,)$ and $\widehat{\boldsymbol{b}_2} = (\frac{3}{5}, \quad \frac{6}{5},)$. Then $\widehat{\boldsymbol{b}_1} + \widehat{\boldsymbol{b}_2} \neq \boldsymbol{b}$.




```
b  = Vector([1,1])
a1 = Vector([1,0])
a2 = Vector([1,2])

A1 = Matrix([a1])
bhat1 = A1 * SEL((~A1)*A1,(~A1*b)).solP

A2 = Matrix([a2])
bhat2 = A2 * SEL((~A2)*A2,(~A2*b)).solP
Sistema([bhat1, bhat2])
```

□

(L-13) Question 4(b) Since $\mathbf{A}$ is invertible, the projection matrix $\mathbf{P} = \mathbf{A}(\mathbf{A}^{\intercal}\mathbf{A})^{-1}\mathbf{A}^{\intercal} = \mathbf{I}$ projects onto all of $\mathbb{R}^2$. Therefore $\widehat{\boldsymbol{b}_1} = \mathbf{P}\boldsymbol{b}_1 = \boldsymbol{b}_1$.

```
A3 = Matrix([a1,a2])
P  = A3*InvMat((~A3)*A3)*(~A3)
bhat3 = P*b
Sistema([bhat1,bhat2,bhat3,P])
```

□

(L-13) Question 5(a) $\mathbf{P}^2 = \mathbf{P}$ and therefore $(\mathbf{I} - \mathbf{P})^2 = (\mathbf{I} - \mathbf{P})(\mathbf{I} - \mathbf{P}) = \mathbf{I} - \mathbf{P}\mathbf{I} - \mathbf{I}\mathbf{P} + \mathbf{P}^2 = \mathbf{I} - \mathbf{P}$.
When $\mathbf{P}$ projects onto the column space of $\mathbf{A}$, $(\mathbf{I} - \mathbf{P})$ projects onto the *left nullspace* of $\mathbf{A}$.

□

(L-13) Question 5(b) $\mathbf{P}^{\intercal} = \mathbf{P}$ and therefore $(\mathbf{I} - \mathbf{P})^{\intercal} = (\mathbf{I}^{\intercal} - \mathbf{P}^{\intercal}) = \mathbf{I} - \mathbf{P}$.

□

(L-13) Question 6(a)

$$\mathbf{P}_1 = \frac{1}{9}\begin{bmatrix} 1 & -2 & -2 \\ -2 & 4 & 4 \\ -2 & 4 & 4 \end{bmatrix}; \qquad \mathbf{P}_2 = \frac{1}{9}\begin{bmatrix} 4 & 4 & -2 \\ 4 & 4 & -2 \\ -2 & -2 & 1 \end{bmatrix}.$$

$\mathbf{P}_1\mathbf{P}_2$ = zero matrix because $\boldsymbol{a}_1 \perp \boldsymbol{a}_2$.

□

(L-13) Question 6(b) $\widehat{\boldsymbol{b}_1} = \frac{1}{9}\begin{pmatrix}1, & -2, & -2,\end{pmatrix}$, $\widehat{\boldsymbol{b}_2} = \frac{1}{9}\begin{pmatrix}4, & 4, & -2,\end{pmatrix}$, $\widehat{\boldsymbol{b}_3} = \frac{1}{9}\begin{pmatrix}4, & -2, & 4,\end{pmatrix}$. Then $\widehat{\boldsymbol{b}_1} + \widehat{\boldsymbol{b}_2} + \widehat{\boldsymbol{b}_3} = \begin{pmatrix}1, & 0, & 0,\end{pmatrix} = \boldsymbol{b}$. Note that $\boldsymbol{a}_3 \perp \boldsymbol{a}_1$ and $\boldsymbol{a}_3 \perp \boldsymbol{a}_2$.

□

(L-13) Question 6(c)

$$\mathbf{P}_1 + \mathbf{P}_2 + \mathbf{P}_3 = \frac{1}{9}\begin{bmatrix} 1 & -2 & -2 \\ -2 & 4 & 4 \\ -2 & 4 & 4 \end{bmatrix} + \frac{1}{9}\begin{bmatrix} 4 & 4 & -2 \\ 4 & 4 & -2 \\ -2 & -2 & 1 \end{bmatrix} + \frac{1}{9}\begin{bmatrix} 4 & -2 & 4 \\ -2 & 1 & -2 \\ 4 & -2 & 4 \end{bmatrix} = \mathbf{I}.$$

□

(L-13) Question 7(a) $\widehat{\boldsymbol{b}_1} = \mathbf{A}_1\left(\mathbf{A}_1^{\intercal}\mathbf{A}_1\right)^{-1}(\mathbf{A}_1^{\intercal})\boldsymbol{b}_1 = \begin{pmatrix}2, & 3, & 0,\end{pmatrix}$ and $\widehat{[1]e} = \begin{pmatrix}0, & 0, & 4,\end{pmatrix}$.

□

(L-13) Question 7(b) $\widehat{\boldsymbol{b}_2} = \mathbf{A}_2\left(\mathbf{A}_2^{\intercal}\mathbf{A}_2\right)^{-1}(\mathbf{A}_2^{\intercal})\boldsymbol{b}_2 = \begin{pmatrix}4, & 4, & 6,\end{pmatrix}$ and $\widehat{\boldsymbol{e}_2} = \begin{pmatrix}0, & 0, & 0,\end{pmatrix}$.

□

(L-13) Question 7(c)

$$\mathbf{P}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ projection onto the } xy \text{ plane.} \qquad \mathbf{P}_2 = \frac{1}{2}\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \left(\mathbf{P}_2\right)^2.$$

□